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2003 Manufacturing Week
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Welcome Students!

New 3M tapes address hidden heat-sink bonding difficulties

By Joseph Ogando, Show Daily Staff

With electronics making their way into so many products today, engineers increasingly find themselves dealing with thermal management dilemmas. And some of the confusion lies with the proliferation of conductive bonding products. Tapes, interface pads, and epoxies all vie for some of the same heat-sink bonding applications.

"There's a lot of misunderstanding about which one to use," says Jeff McCutcheon, a senior technical specialist with 3M (St. Paul, MN).



So which solution is right? 3M, which supplies all three of these thermal management products, recently conducted research that at least clears up some of the misconceptions about thermally conductive adhesive tapes for heat sink bonding.

"We evaluated typical heat sink and chip package configurations," explains McCutcheon. One of these tests, for example, involves the measurement of heat sink and chip package surface contours.

"We found that many have contours and surface features that people are not fully aware of," McCutcheon says. In a representative test of a customer's heat sink design, 3M discovered a seven-mil hill in on the package. "That's pretty hard to detect when you run your finger over it," he adds. But it's enough to decrease the crucial contact between the heat sink, tape, and package—and thus lower thermal impedance.

3M has likewise run tests to characterize the adhesive wetout—or amount of adhesive contact between the tape and the bonding surfaces. This test employs a camera and special software to measure the amount of wetout of tapes applied to a glass slide. A vitally important attribute, the amount of wetout plays a key role in the thermal and mechanical performance of the heat sink system, McCutcheon notes.

Yet 3M discovered that wetout can be lacking. McCutcheon cites 70% wetout as a good minimum to get adequate thermal and mechanical performance in most applications. Yet 3M's glass-slide tests revealed as little as 20% surface wetout for some real-world thermal management designs.

3M recently came out with a new line of thermally conductive adhesive transfer tapes that spring from all this on-going research. These 8800 Series tapes—which come in 5-, 10-, 15-, and 20-mil versions—have been optimized to provide a better bond between heat sinks and chip packages. They achieve this goal in a couple of ways. For one, the tapes are "more conformable than previous products," McCutcheon reports.

This conformability comes partially from the inclusion of relatively thick tapes in this line. "Thicker isn't always considered desirable in electronics packaging," McCutcheon concedes. But in this case, thicker tapes can help them conform to surface irregularities. For another, the pure adhesive construction of these tapes also makes them more conformable—since they have no stiff carrier to interfere with gap filling, McCutcheon

(Continued on page 7)

Also On Tap Today

9:15 a.m.: Design for No Assembly. **1A.10**

10:45 a.m.: Optimizing Your Design. **1A.11**

1:30 p.m.: Advanced Nanomanufacturing Technologies. **T2.12**

1:30 p.m.: What is Lean Manufacturing and How do We Make it Lean Manufacturing? **T3.12**

NMW Exhibit Hours: 10 a.m.-3 p.m.

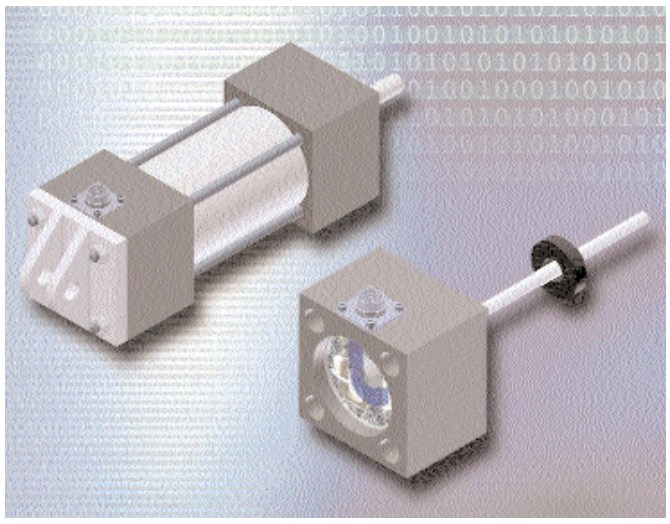
Electronics and NDES: Something for everyone

By Rick DeMeis, Show Daily Staff

Putting an engineer on the show floor at the National Design Engineering Show (NDES) is like turning a kid loose in a candy shop. Take a twirl around this year's exhibits and you'll come away with something new and useful down every aisle. Here's what we've noted particularly in electronics that you may want to check out in some of the booths.

Swing by Booth 5348 and you'll find MTS Sensors Division's compact Temposonics® L Series LD4 position sensor. The company says it has the shortest installed height available at 28.5 mm. To get into tight places, engineers used a short sensing element and an overlapping electronics module to allow the complete sensor to be mounted within an actuator cylinder endcap—which in turn protects the sensor. The small operating envelope allows easy installation by simply fastening the sensor to the connector. And the complete cylinder needn't be disassembled for installation or maintenance.

Over at Cherry Electrical Products, in Booth 6554, are the company's latest CR Series power rocker switches, capable of switching 20A. These switches come in lighted and non-lit versions and a variety of colors. In their splash-resistant versions, the switches feature a housing configuration that channels liquid or particles away from internal parts.



The MTS Temposonics LD4 position sensor is targeted at space limited applications and features eased installation.

The folks at Phoenix Contact, Booth 4701, are showing off the Vario-face PLC-V8L adapter that provides relay connectivity to controllers. Dedicated Controller System Cabling, with ribbon cable plug connectors, facilitates connecting relay interfaces to PLC and DCS output cards—for what the company says is a time saving of upwards of 60%. The pluggable relays are replaceable without disconnecting any wiring. And the insulated bridge plugs in to permit quickly "commoning" relays without time-consuming daisy chaining.

E-T-A Circuit Breakers, in Booth 6345, has its DIN-rail ESS20 breaker that selectively protects individual circuits connected to a common 24V dc switch-mode power supply. Switchable to either a 3 or 6A rating, the breaker targets the problem of having an overload in a single circuit trigger the shutdown of output to all circuits it supplies—crippling for entire industrial processes. Functioning in this manner, the device also eliminates "hiccoughs" of quickly cycling on and off and thus not supplying adequate voltage—generating scrap materials and endangering workers. The ESS20 allows quick identification of the faulty circuit while limiting the spread of any disruption.

Head on over to the Pliz Automation Safety display, Booth 4536, to look at the PNOZmulti, a modular safety system. The system consists of the company's PNOZm 1p basic device with a range of enhancement modules, and provides four to 14 safety functions. Configuration software provides flexibility with

(Continued on page 7)

How well do ISVs support Microsoft's technology?

By Joy LePree, Show Daily Staff

As a National Manufacturing Week show attendee, chances are you'll walk past scads of booths displaying software running on one Microsoft system or another. But how well do these independent software vendors—or ISVs in software-industry speak—actually leverage Microsoft technologies?

"It might be difficult to tell which ISV offers a more sophisticated usage of Microsoft technology just by walking past the booth," says Don Richardson, director of Manufacturing Industry Solutions for Microsoft, (Redmond, WA). "But there are a few things you can look at to determine how well they leverage our technology."

The first thing to notice, Richardson says, is the browser. "Look at the browser to determine the richness of the user experience. You want the Web site to be alive with graphics and spread sheet capabilities," he says. "Also, find out how well the ISV supports the whole solution stack of Microsoft technologies. The more they leverage our platform, the more time they have to focus on their own value-added expertise."



Don Richardson

While talking to ISVs at the show, be sure to check out some of the newest .NET- and Windows XP-based applications on display.

One example you'll see is SYSPRO 6.0, the latest enterprise software offering running on .NET from SYSPRO, a Costa Mesa, Calif.-

(Continued on page 7)

Sefar exhibiting precision filtration materials

Sefar America Inc. (Depew, NY) is exhibiting a variety of products at the National Design Engineering Show, Booth 6061. The products include precision woven plastic and wire screens and filter media with opening sizes ranging from <1 micron to more than 5,000 microns. Polymers include polyester, polyamide, polypropylene, PEEKä, fluoroplastics, and alloys include stainless steel, aluminum, copper, and titanium.

Sefar America is a supplier of precision woven filtration materials. State-of-the-art manufacturing facilities in Switzerland, France, and the U.S. use proprietary processes to provide the most consistent, repeatable, and reliable woven screening and filtration media available.

Medifab®, Pecap®, Nitex® and Propyltex® are recognized as the industry standards for excellence in synthetic woven fabrics. Medifab® synthetic fabrics have been developed specifically for healthcare customers who need clean, pyrogen-free products. Customized surface tension modifications and hydrophilic or hydrophobic coatings are also available. Synthetics can be dyed to match custom colors.

Applications include filtration, sound attenuation, EMI/RFI shielding, sieving, particle classification, coalescing filtration, composite laminate reinforcement, membrane spacers and protective screens for microphones or speakers.

Off-the-shelf roll goods can be delivered quickly, or custom components can be cleanroom fabricated in ISO9001 certified facilities. Conversion capabilities include stamping, slitting, heat sealing, ultrasonic and RF welding, laser cutting and pleating–all backed by local expert technical support.

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Ticona features engineering plastics

Engineering thermoplastics allow designers and manufacturers to solve tough problems to make their products work in challenging situations. Ticona (Summit, NJ), a leading global supplier of engineering polymers, demonstrates the many ways these plastics are used at booth 6328 at the National Engineering Design Show.

Under the theme, "Making Performance Dreams a Reality," the booth showcases the "realities" that Ticona's plastics help make possible in automotive, industrial, and consumer products.

The booth also highlights current and potential applications in medical devices, packaging, and new technologies, such as high-performance nonwovens and fuel cell components.

These end uses draw on the company's eight engineering thermoplastic lines to gain advanced mechanical, electrical, chemical, thermal and other properties. Ticona's product lines include:

Celcon® and Hostaform® acetal copolymer (POM)

Celanex® PBT, Impet® PET, Riteflex® and Vandar® thermoplastic polyesters

GUR® ultra-high molecular weight polyethylene (UHMW-PE)

Vectra® and Vectran® liquid crystal polymers (LCP)

Fortron® polyphenylene sulfide (PPS)

Topas® cyclic olefin copolymer (COC)

Celstran® and Compel® long fiber reinforced thermoplastics (LFRT)

Celanese® nylon 6/6

Specific existing and potential applications featured at the Ticona booth include:

Washing machine transmissions and dishwasher parts made of Celcon acetal, that creates cost-effective components that are quieter, lighter, tougher and more resistant to chemicals and wear than metal parts.

An automotive fuel rail made from Fortron PPS that weighs 25 percent less and costs 30 percent less than comparable steel rails. This single-piece, injection-molded unit greatly simplifies production compared to the formed,

Matte-finish on dashboard controls and other interior parts using a new low-gloss grade of Celcon® acetal. This grade not only creates a soft touch and stylish look, but also can match other interior components without painting. It also resists scratches, cleaning agents and UV radiation.

Medical devices based on our medical technology (MT™) resins, including elements in dry powder inhalers made of Celanex® PBT and Celcon acetal. These FDA-compliant polymers provide high dimensional and thermal stability and chemical resistance. All Ticona MT resins have low extractable levels, are USP Class VI compliant and have a consistent and high purity level.

Drug packaging and delivery devices, based on Topas® COC, range from drug blister packs to syringes and vials. Topas COC offers excellent transparency, so it is often used to replace glass. It also is highly pure, rigid and resists acids, bases and polar solvents.

Modularization, such as automotive front-end modules and door modules, using Celstran® LFRT. This material is stiff and tough, dimensionally stable, and has low warpage and a good surface appearance. It allows engineers to replace metals, traditional short-fiber reinforced thermoplastics and sheet molding compounds, and deliver lightweight, integrated components.

For further information about Ticona and Celanese, visit www.ticona.com and www.celanese.com.

Performance, simplicity, drive test tools

By Rick DeMeis, Show Daily Staff

Paul Heydron is Product Planning Manager, Fluke Industrial Group for the Fluke Corporation (Everett, WA). Here he talks about the various trends he sees in the electronics industry during a conversation with Rick DeMeis.

Show Daily: What's new in hand-held test instrumentation?

Heydron: There are a number of trends driving new product design in the hand-held test tool market: safety and ruggedness, ergonomics, matching the instrument to the environment in which it is being used, simplified user interfaces, non-contact measurement whenever possible, greater reliability, and, finally, expanded capabilities such as data logging.

Q: What is on the horizon in the next five years?

A: Greater capability, which will have to be balanced by the consumer demand for simplicity. This will be driven by firmware. An example is an algorithm to determine inrush current that is accessed by a touch of a button. In the past, inrush was determined by the electrician doing some complex formulas in his head.

Q: Is the Internet having any impact on the way you are doing business?

A: The Internet is a powerful tool that allows us to communicate with customers quickly and efficiently. We use it to conduct research, to "test" product ideas, and to monitor acceptance to new products. Perhaps most importantly it allows us to publish information about how to use our products that customers can read and print out. By studying the most frequently accessed areas of our web page we get a good idea of what people are interested in, which helps in product development and design.

Q: What is the effect of the Internet on the way your customers work?

A: The Internet has driven a rather surprising change in the way many of our customers work. Ten years ago a manufacturing company might have had more than one network, for example, one IT/MIS network and a completely separate production network. Over the last 10 years, the Internet has been driving down connectivity costs, particularly using Ethernet and TCP/IP. As costs went down and data requirements went up, companies began connecting the IT/MIS and production systems networks. The most cost-effective way to do this was to eliminate proprietary networks and drive Ethernet as far down into the production system as possible. In many cases this means that control and order systems are essentially on the same network, with both types of information visible from web pages. So how does this impact the traditional production system technician? In many cases an engineer or technician now has to be conversant with networks as well as control systems. We find people buying ScopeMeters to troubleshoot difficult network problems on the shop floor and purchasing networks diagnostic tools to use along with their digital multimeter (DMM).

Q: How are your customers' needs changing?

A: They have less of the following: time to do their job, training, and patience for learning how to use a new piece of equipment. Again, the challenge is working to balance the capabilities of the tool, making sure it can do everything the user requires, with the need for a highly intuitive interface, which the customer can understand and begin using without reading the manual. This requires the product designers, from the standpoint of component and software engineering, and industrial design, to go out and invest time in talking with customers, watching how they work, and testing new ideas with them before starting production.

Exact Software demonstrating Macola ES ERP system

Exact Software announced Macola ES, the latest and most revolutionary version of the company's robust ERP solution. Exact has eclipsed traditional ERP solutions by introducing new functionality for business process management. Through the use of this new functionality, mid-market organizations and large enterprises will now have the ability to leverage a fully integrated front- and back-office.

As the focal point of Exact's corporate news at National Manufacturing Week, Macola ES will take center stage at Exact's booth 617 in the Microsoft Pavilion. Visitors to the Exact booth will have the opportunity to view Macola ES in action through live demos and Exact's staff will be on-hand to provide insight on how Macola ES can have an immediate impact on improving organizational workflow and create new levels of operational efficiency.

The product demonstrations and business insight won't be limited to just Macola ES. In addition to Macola ES, Exact's consulting team will also be demonstrating the business value of e-Synergy, Exact's enterprise relationship management solution, which made its debut in North America in October. Organizations with an ERP solution such as Microsoft Great Plains already up and running, will have a particular interest in learning more about e-Synergy. Capable of integrating with a number of industry-leading back-office solutions, e-Synergy helps organizations automate traditional front-offices business processes to enable greater control of their businesses.

Free commuter rail rides available under McCormick Place

By Jim Montague, Show Daily Staff

To avoid waiting in long lines for taxis and shuttle buses at National Manufacturing Week 2003, visitors can use McCormick Place's own commuter rail stop with free, five-minute train rides to and from two stations that are much closer to downtown Chicago.

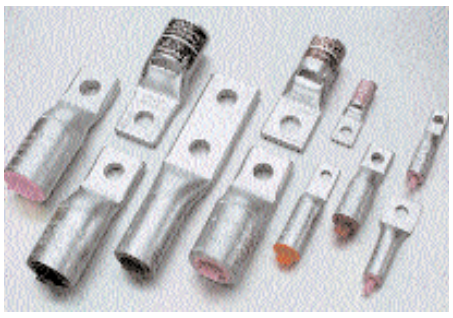
Located beneath the huge convention center's North building, this subterranean station is easily reachable from McCormick Place's main concourse. If you're walking west in this main hallway between the North and South buildings, towards those taxi and bus lines outside the main exit, you can instead head toward the right side of the hall to find the stairway down to the platform. There is usually an information booth marking the stairs, with schedules and free train passes for NMW 2003 visitors, located halfway between the exhibit halls and the main exit doors.

Every few minutes during rush hour, commuter trains run north a mile or two to Chicago's Monroe Street and Randolph Street stations, which are both located a block east of Michigan Avenue and three blocks east of the Loop area. The one-way trip takes five minutes, and drops riders off where taxis are often more plentiful, lines are short or non-existent, and driving times and fares to nearby hotel are significantly less than from the show.

Likewise, there are just as many trains running south to McCormick Place in the morning, which makes reversing this strategy an equally good way to travel to the show. The free passes are good for all four days of the show.

FCI-BURNDY showcases new E-Line connectors at Plant Engineering Show

FCI-BURNDY (Manchester, NH), a leading manufacturer and provider of connector solutions to the industrial, energy, and automotive industries, introduced its new E-Line Compression Terminal Connectors and E-Line Maintenance Kits at Booth 2445 at the National Plant Engineering & Management Show.



FCI-BURNDY introduced its E-Line Compression Terminal Connectors.

The E-Line Connectors are specifically designed for applications in which an electrical connector is terminated in a confined area. Featuring a special, narrow tongue lug, FCI's E-Line Connectors and Maintenance Kits are ideal for space-restrictive OEM and repair and maintenance service applications, including: switchboards, panel boards, motor control centers, switchgear, circuit breakers, switch products, control products, and transformers.

E-Line Compression Terminal Connectors are available in copper or aluminum, and are UL listed 90° C, 600 volt to 35kv. E-Line Maintenance Kits come complete with either copper or aluminum E-Line Compression Terminal Connectors and Heat Shrink Tubing for proper terminations.

Mid-sized companies benefit from advanced warehousing technology

Adonix recently enhanced its flagship product, Adonix X3–a robust enterprise resource planning (ERP) solution for mid-sized companies–to strengthen its warehouse management and data collection capabilities.

"The new features are designed to address the needs of mid-sized companies that have demanding warehouse requirements. We believe that customers that integrate warehouse management processes into their ERP system will see a significant cost savings and gain a major advantage over their competitors," said Alex Attal, CEO of Adonix North America. Adonix X3's new warehousing features include:

- Directed Putaway to appropriate storage locations by radio-frequency (RF) scanning according to factors such as item velocity, weight, dimensions and consolidation rules
- Palletized Picking and Consolidated Shipping, including RF verification and capture of pallet and shipment identifications
- Pick Planning and Scheduling, including directed picking for single shipments or groups of shipments, based on warehouse location
- Warehouse Labor Tracking to provide labor efficiency and utilization reporting to management

The new features set Adonix X3 apart from most current manufacturing-oriented ERP products in the market, which still do not innately support automated data collection or have the ability to manage and fulfill an order that might involve increasingly required 'postponed manufacturing.'

Additionally, Adonix offers a complete "one-stop-shop" solution. Besides providing Adonix X3 ERP software, integrated data collection, barcode label management, and full implementation services, the company offers data collection equipment manufactured by leading hardware vendors. The result is a completely integrated solution designed to provide companies with a rapid return on investment through greater accuracy, faster cycle times, and lower costs.

NMW Sponsor Message

EDS' PLM technology offers complete solutions

Born from EDS' acquisitions of Unigraphics/SDRC, and focused under the leadership of executives from both companies, EDS PLM Solutions represents a comprehensive approach to extended enterprise collaboration that enables all participants in a product lifecycle to work in concert as organizations bring products to market and support the customer base.

EDS is the only company capable of providing the complete set of PLM solutions needed to turn a company's product lifecycle process into a distinct and sustainable competitive advantage.

EDS PLM Solutions offers the most comprehensive suite of product lifecycle management solutions in the industry. Combined with leading global information technology (IT) services, EDS is the PLM solution provider of choice. EDS PLM Solutions' software includes product development, enterprise collaboration, data management, and factory and manufacturing planning.

Unigraphics® NX and I-deas® NX Series

EDS PLM Solutions' NX offerings provide an integrated and comprehensive family of total product engineering solutions that enable the user to digitally create, capture, and analyze 3D product definitions. These applications are used by many of the world's leading manufacturers to perform conceptual, industrial and detailed mechanical design along with engineering simulation and digital manufacturing.

Teamcenter®

Teamcenter is a suite of offerings that enable extended enterprises to capture, manage, access, integrate, and leverage diverse types of product information in a web-native environment, including product requirements, project data, engineering design data, parts/documents, and product configurations. EDS PLM Solutions has developed industry-specific Teamcenter solutions in the automotive, aerospace and defense, high-tech electronics and consumer packaged goods markets.

Solid Edge®

Solid Edge provides a family of capabilities that enable organizations to create and capture 3D product definitions for mechanical design, drafting and assemblies with special emphasis on lower cost-of-ownership and easy-to-use upgrade paths from traditional 2D-design.

E-factory™

E-factory is a manufacturing information management solution coupled with a suite of digital manufacturing applications. It is an open system designed to manage data from a variety of manufacturing applications and provides a platform for managing the pre-production manufacturing-planning phase of a product's lifecycle.

The success does not stop with these products. EDS PLM Solutions is constantly expanding its domain expertise and reach in discrete and non-traditional industries, where collaboration and faster to market are driving factors. With over two million seats in operation, EDS PLM Solutions products are becoming the corporate standard for PLM in companies of all sizes around the world.

Wavetek Meterman unveils digital multimeters



Visit Wavetek Meterman in Booth 2212 at the Plant Engineering MRO Show to view its full array of test equipment as well as its new product offering of digital multimeters, the XR Series.

Wavetek Meterman® Test Tools' XR Series Digital Multimeters are designed specifically for today's electronics test, field service, and industrial applications.

The five-product Meterman XR Series is distinguished by a new ergonomic design. Each meter has a sleek, curved shape integrating a functional Magne-Grip® holster with built-in magnetic hanging strap plus an easy-access battery door.

In designing the XR Series DMMs, a considerable amount of time and expense was devoted to listening to customer feedback. This process led to an entirely new approach combining function and convenience by integrating normally extra-cost accessories such as the Magne-Grip® holster and molded probe holders into the basic design form, the company reports.

The new-product strategy of including normally extra-cost items in the standard package shows how versatile DMMs can be. As an example, for better viewing in dim light situations, some XR models offer an exclusive new bright, blue-neon backlight display. Another unit, such as the entry-level 30XR product, offers a built-in non-contact voltage detector that can safely signal live electrical circuits without any additional options.

Each unit in the series is designed with different features and functions to match the discrete needs of electricians, industrial plant managers, electronics technicians, field service troubleshooters, or the home do-it-yourselfers:

- 30XR Digital Multimeter with non-contact voltage detector, $59.99
- 33XR Digital Multimeter with temperature, frequency capacitance, $89.95
- 34XR Autoranging Digital Multimeter with true-rms, backlight, temperature, frequency, capacitance, and an analog bargraph plus 3 year warranty, $109.95
- 37XR Autoranging Digital Multimeter with true-rms, backlight, component test functions, 0.1% accuracy, CAT II 1000 V, CAT III 600 V capability, $129.95
- 38XR Autoranging Digital Multimeter with true-rms, backlight, CAT III 1000 V, CAT IV 600 V range and RS232 capability with optional software, $149.95

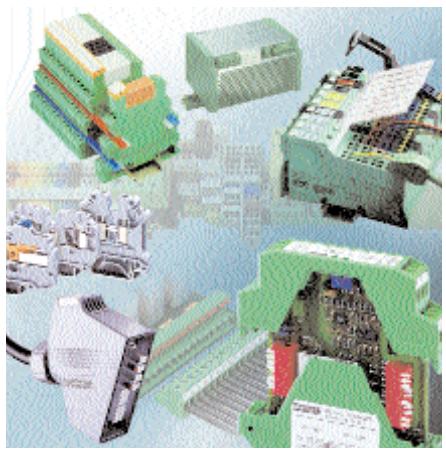
Phoenix Contact exhibits connection solutions

Phoenix Contact Inc., a leader in the manufacture of industrial connection and automation technology, is exhibiting a variety of product innovations at the Industrial Automation Show (Booth #4701) at National Manufacturing Week.

The company's 50' x 50' booth will highlight products featuring control panel connection technology, Industrial Ethernet, embedded control and I/O solutions, and power connections. Product displays will emphasize some of the ways Phoenix Contact components can solve design and installation problems and create new opportunities for efficiency.

Exhibit attendees will learn how they can optimize their time and space when using the company's product offerings. Live product panel demonstrations will allow show attendees to see products in action and enable them to see the product applicability for a number of industry applications.

For more than 75 years, Phoenix Contact has manufactured products that connect or control industrial applications in a wide variety of industries, including industrial controls, material handling and packaging, machine tool, water/wastewater, and utilities.



Phoenix Contact will exhibit numerous products at the Industrial Automation Show.

With a line-up of more than 20,000 different products and dozens of new innovations created each year, Phoenix Contact remains committed to meeting customers' current and future needs.

Phoenix Contact is registered to ISO 9001: 2000 - Quality systems model for quality assurance in design, development, production, installation, and servicing.

How well do ISVs support Microsoft's technology?

(Continued from page 1)

based vendor of enterprise systems and supply chain management solutions.

"SYSPRO e.net solutions leverage the Microsoft .NET framework as the foundation for delivering a new generation of customizable, Web-based applications and services using the XML standard," says Brian Stein, president of SYSPRO. "The benefit of using this technology is that it's a cost-effective way for our customers to integrate best-of-breed applications, maximize business-to-business trading, and leverage wireless connectivity. By capitalizing on this technology, the functionality of our core solution can be enhanced to match individual business requirements."

And, Microsoft XP technologies are used extensively in supervisory control software from GE Fanuc Automation's Intellution business to enhance the day-to-day activities of managers and other personnel in manufacturing plants.

"We use the built-in security features of Windows NT, 2000, and XP to simplify and reinforce the security within our own applications—specifically those with functionality designed to meet the demands of the Food and Drug Administration and regulation 21 CFR Part 11," says Roy Kok, product marketing director of supervisory applications.

"Our iFIX 3.0 product ties its security model in with that of the underlying Windows operating system. This makes it easy to set-up and administer and eliminates the headaches, expenses, and inefficiencies of keeping a separate username/password database elsewhere in the system."

Since manufacturing is one of the industries where Microsoft has the highest rate of penetration, it is likely that show attendees will come across these and many other enterprise and plant floor solutions running on Microsoft technologies.

"As you're walking around, just keep in mind that while Microsoft brings an excellent platform, the ISV should focus on the vertical of their own expertise," says Richardson.

New 3M tapes address bonding difficulties

(Continued from page 1)

explains. These improvements to the conformability all contribute to surface wetout values that 3M places well in excess of 70%.

The tapes offer enhanced strength as well, for better results in battery of mechanical tests that electronic components go through. Some of the enhancement comes from the improved wetout, but the 8800 Series tapes feature a redesigned adhesive that plays a role here too. According to McCutcheon, the new tapes offer a 50% strength improvement in torque tests—which involve trying to twist the heat sink off the chip package. And customers have reported similar improvements in shock and vibration tests as well, McCutcheon adds.

Finally, the revamped adhesive allows this acrylic tape to bond to plastic, ceramic, and metal substrates. "Some tapes need a separate layer of expensive silicon adhesive to adhere to all these substrates," McCutcheon says.

For more information, as well as detailed mechanical and thermal data for the 8800 Series, visit www.3m.com/conductives or contact Jeff McCutcheon at jwmccutcheon@mmm.com

NMW Sponsor Message

2003 will be Year Of Design Interoperability

By Ilya Mirman
Vice President of Marketing
SolidWorks Corporation

One of the year's most important trends focuses on the front end of the manufacturing process: 2003 is shaping up to be the Year of Design Interoperability. Companies are looking to unshackle themselves from clunky, legacy computer-aided design (CAD) tools that fail to interoperate with other CAD tools.

Design engineers are now collaborating across workgroups, divisions, and corporate boundaries without major software installs. They can communicate designs, exchange design data, find the right partners, and incorporate standard parts in a design without having to create part models from scratch.

For example, we at SolidWorks have found a way to exchange drawings and 3-D CAD models by e-mail in a way that lets the recipient view and comment on them with a mark-up pen regardless of his or her CAD package. In fact, the recipient doesn't need a CAD package at all. We call this product eDrawings Professional.

To foster exchange of actual design data, we have built into our CAD system the capability to read other vendors' native files. The latest version of our flagship 3D CAD software, SolidWorks® 2003, is the leader in interoperability and can read and write more native file formats than any other CAD system on the market today. And while some of our competitors use encryption to prevent other CAD systems from reading their files, we work hard to deliver interoperability.

Today, an important trend for CAD software vendors and others vying to be manufacturers' business partners is in helping these manufacturing customers identify suppliers with state-of-the-art CAD interoperability tools in order to completely eliminate concerns about data interoperability. Such compatibility helps companies turn products around faster, since there's no need to redesign parts from scratch or fix ("heal") imported design files due to interoperable CAD systems. Scrap rates are lower because data translation problems are eliminated, and higher first-try success rates result.

The SolidWorks Manufacturing Network (www.solidworks.com/swmn) directly addresses these concerns. The network provides a free Internet directory for quickly finding fellow SolidWorks software users who provide a broad range of contract design and manufacturing services representing every step in bringing a design to market. These include design, machining, mold making, rapid prototyping, finite element analysis, and many others.

Finally, we've used the Internet to offer 3D ContentCentralSM, a free online directory of downloadable 3D CAD parts from leading standard component manufacturers.

3D ContentCentral lets companies download drawings and models of parts to check form, fit, and function. By letting people reuse instead of redraw an existing design, the design process becomes more efficient. The customer gets their product to market faster, without taking hours to design a standard part from scratch. Soon, the days of redesigning the same part multiple times along the supply chain will be gone.

Design interoperability like this compounds the efficiencies that 3D brought to CAD, and that CAD brought to design in the first place. As a result, engineers can more quickly design better products for significantly improved customer satisfaction.

Electronics and NDES: Something for everyone

(Continued from page 1)

simplicity—the company says no special knowledge is needed to configure safety functions such as E-STOP switches, safety doors, light curtains, etc. or monitoring operations.

You can get a good feeling in the Sensor Products booth (6033) with its Tactilusâ electronic tactile force and pressure-indicating sensor. Useful for pressures from 0.01-9,000 psi, the sensor permits monitoring how force is distributed between two contacting surfaces in real time. Tactilus is Windows-based and consists of an electronic sensor element "skin," an interface controller, and software. The flexible skin consists of sensing points spaced as close as 1 mm apart, and it can collect data as fast as 60,000 points per second.

Dresser Instruments, in Booth 7418, is displaying the Ashcroft Type D1005PS digital pressure gauge. Terminal point accuracy is given as 0.5% full scale, including linearity, repeatability, and hysteresis. The eleven ranges available go from vacuum to 9,999 psi and standard features include a peak hold for maximum pressure indication.

Smooth operator interfaces are the order of the day down in the IDEC Corp. booth (4316). Here the latest additions to the HG family, HG3F and HG4F, sport 256-color TFT LCD touchscreens and replaceable backlights. These large displays (10.4 and 12.1 inches diagonal, respectively) weigh only 6.6 lbs. Other features include 8 Mbyte of memory, alarm logging, and trending capability.

Sealcon, in Booth 5947, is showcasing its M23 Circular Power Connector, which comes in straight, elbow, the proprietary Swivel Snap Elbow, and any mating panel connector (receptacle) for threaded or single holes. The liquid tight (NEMA 4X (IP67)) connector has integral cable strain relief. It accommodates cables from 7- to 14-mm diameter and is rated from 8-28A and 300 or 800 V/pin.

This abbreviated "tour" of the NDES floor only scratches the surface of new electronics products available. Cruising the aisles and kicking the tires of what's on display in each yourself will reveal even more ideas you can take back to your job.

Design Engineering Show Products

Stainless steel valve

The Model 8273, a new valve in the Series 8 family, is a two-way diaphragm unit available in stainless steel. It has a click-on coil that can be exchanged to accommodate various coil voltages by pulling back on a metal clip. The unit operates at 1.5 to 232 psi pressure. Ambient temperatures range between 14º and 122ºF, and fluid temperatures are between 14º and 194ºF. EPDM and fluorocarbon seals are also available.

Norgren-Kip Fluid Controls
Booth 5340 www.kipinc.com

PLM Solutions

For small-, medium-sized businesses CATIA Version 5 Release 10, developed by both the listed company and IBM, features extended end-to-end processes coverage for OEMs and a CATIA-ENOVIA Work Package Exchange for supply chain collaboration. Made with a V5 architecture, it allows customers to create process-driven applications, with more efficient collaborative engineering.

Dassault Systemes of America
Booth 5917 www.3ds.com

PDM Solution

PDMWorks™, the company's PDM solution, enables users to search for, revise, and store product design information, to avoid being lost or overwritten. Allowing people to work in a secure environment, the PDM solution offers automated revision control, and a history of design activities. Other new features include support for the company's configurations, support for custom file property manipulation, support of lifecycle status labels, and SolidWorks Toolbox compatibility.

SolidWorks Corp.
Booth 6501
www.solidworks.com

Servo drives

PCE800 Series servo drives are DSP-based units featuring 480 Vac operation. They have a footprint up to 40 percent smaller than comparable drives and are small enough to fit into a 10-in. cabinet. They are designed to keep complex mechanical systems running cost-effectively.

Pacific Scientific, Div. Danaher Motion Co.
Booth 5112 www.danahermotion.com

Twist clamp cylinder

The Twist Clamp cylinder incorporates an internal mechanism to produce rotary piston motion throughout stroke. It's based on the EF1 cylinder. By attaching a clamp arm, the actuator becomes a clamping mechanism for light-duty holding applications. The rotary motion rotates the clamp arm out of the work area for easy loading and unloading of parts. The Twist Clamp is available in four bore sizes, 25, 40, 50, and 63 mm. Clamping force of up to 600 lbs can be achieved.

Bimba Manufacturing Co.
Booth 5136 www.bimba.com

Flow control systems

Pressure, flow, and measurement control systems can be custom assembled to reportedly save time and money on engineered components. A broad range of standard valves, regulators and pressure control components is available for controls from vacuum to 30,000 psig. Simplify existing systems or create new ones using services like direct engineering support and custom software. Applications include contractor, government, military and international standards in materials like 316 stainless steel, Monel and Hastalloy®.

Tescom - Industrial Controls Div.
Booth 7314 www.tescom.com

Linear actuator

The LA12 linear actuator is reportedly a compact and lightweight, PLC-ready version ideal for factory automation and industrial equipment controls. Built-in electronics including relays, limit switches and positioning feedback allow designers to use fewer components and spend less time wiring, mounting and testing. The LA12 is recommended as an alternative to pneumatic systems, gear motors and standard linear actuation.

Linak US Inc.
Booth 5460 www.linak-us.com

PTC spreads 'Wildfire'

PTC offers its Pro/ENGINEER Wildfire product design and development software. Pro/ENGINEER Wildfire reportedly brings gains in simplicity, power, and connectivity. Added simplicity comes from an entirely new user interface, and flexible workflow. Users have the power to create product designs without experiencing limitations in developing the part, the product, or the process, says PTC, due to its Granite interoperability kernel, as well as via other enhancements. The software also incorporates Web services into its underlying architecture to improve connectivity.

PTC
Booth 427 www.ptc.com

Positioning table

Available with ball or crossed roller elements for loads of 225 or 450 lbs., respectively. Features include smooth motion, rolled lead screw, and accuracy and repeatability to 0.0005 in. on 6-in. total range. Screw reached speeds of 300-800 rpm and acceleration to 12 in/sec, depending on lead screw selection. Tables are supplied with NEMA 23 motor mount and coupling or hand crank.

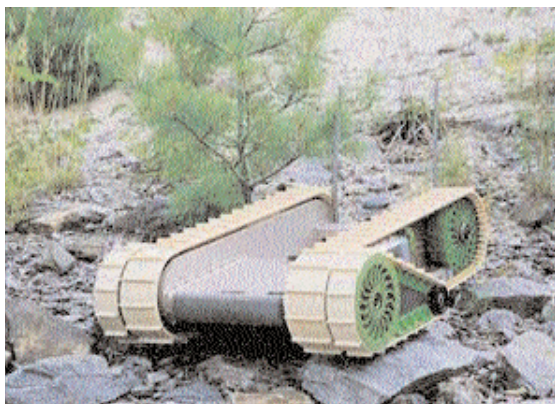
PIC Design
Booth 6526 www.pic-design.com

Stepper motor

Mdrive34 Motion Control is a NEMA 34 high torque stepper motor 24-75-volt microstepping drive with a motion controller and optional internal encoder. Features include optional closed loop control, stall detection, position maintenance and find index mark. Motors include single or double shaft with control knob. Rotary version in motor lengths 24, 31 and 47 are available. Standard Mdrive 34 Motion Control units include 4 I/O lines, and 5 to 24V. Interfaces use 12-in. flying leads.

Intelligent Motion Systems Inc.
Booth 7726 www.imshome.com

(Continued on page 11)



The Packbot from iRobot Corp. will be wandering around the *Design News* booth during the National Design Engineering Show.

Design News booth showcases big and small technology

By Debra Bulkeley, Show Daily Staff

A backpack-sized robot, similar to those that found victims in the aftermath of the World Trade Center attack on 9/11, will roam booth 5126 at the National Design Engineering Show (NDES).

The Packbot from iRobot Corp. (Burlington, MA) actually looks like a small tractor-it moves along on treads, which it will do in the *Design News* booth at the NDES.

NDES attendees are invited to experience the interactive exhibits at the *Design News* booth that are both entertaining and educational.

The remote-control operated Packbot won't be the only robot in the area–robots designed by students from FIRST (For Inspiration and Recognition of Science and Technology) will also grace the exhibit space.

FIRST, a program founded by inventor and technology visionary Dean Kamen, inspires pre-college students to appreciate science and technology. It designs technology competition programs to build self-confidence, knowledge, and life skills while motivating young people to pursue careers in technology.

On the other end of the "things that move" spectrum, visitors will be able to see Volvo's new entry into the Sport Utility Vehicle market, the XC-90. While the SUV will clearly not be moving around the *Design News* booth like the Packbot, there will still be plenty for visitors to admire and see. The seven-seat XC-90 touts safety technology that includes a system of gyroscopes that help prevent "rollovers," as well as numerous other safety systems. The vehicle will be available in September.

If hydraulics is your thing, then Turbine Technologies' (Chetek, WI) miniature steam power plant should interest you. The company's "pump lab" will illustrate different aspects of hydraulics via a "clean system" so that attendees can see how the system operates. Controlled by National Instruments' LabVIEW, company representatives can set up different situations and attendees can clearly see results.

The new tablet PC from Hewlett-Packard (Houston, TX) will also be on display, and attendees can see first-hand the versatility of the technology that lets you enter data either through the writing tablet or keyboard.

Paula Porter, *Design News* senior web editor, will also be at the *Design News* booth demonstrating the publication's on-line products.

Great Expectations for National Manufacturing Week 2003



Six shows in one, more than 1,100 exhibitors, and 100 technical conference sessions will headline the National Industrial Automation Show and National Manufacturing Week, March 3-6, at Chicago's McCormick Place.

By Jim Montague, Show Daily Staff

Nobody has time to waste these days. Control and automation professionals need useful ideas now and demand practical solutions immediately.

To provide these crucial innovations, the National Industrial Automation Show (NIAS) at National Manufacturing Week (NMW) 2003, 10 a.m. to 5 p.m. on March 3-6 at McCormick Place in Chicago, has been reinvented and reorganized.

NMW 2003 will include more than 1,100 exhibitors and special events in approximately 310,000-square-feet of exhibition space. It will also feature more than 100 sessions in a newly expanded technical conference with six tracks designed to parallel four of NMW's six shows. Organizers say they hope the event will attract about 30,000 visitors during its four-day run.

Content and layout

"Content is the big difference this year. There will be a lot for attendees to do and sink their teeth into," says John Stuttard, industry vice president for NMW, Reed Exhibition Cos. (Norwalk, CT).

"Sure, we have all our exhibitors, but we're going beyond that to give [show attendees] access to industry leaders, who can also provide useful information and intelligence," he adds. "Each of the six shows will have its own state-of-the-industry address."

For example, Bill Swanton of AMR Research will moderate the automation show's free session, "Consulting, Out-

(Continued on page 17)

NMW Sponsor Message

Manufacturing: Productivity Powerhouse

By Jerry Jasinowski
President, National Association of Manufacturers

National Manufacturing Week at McCormick Place is one of the few places you can go to see and appreciate the tremendous contributions manufacturing makes to our country's economic strength. Inventing and harnessing technology, boosting productivity to new heights, and creating whole new processes that transform production are all on vivid display this week.

These dynamic forces shape not only today's modern manufacturing, but also the overall U.S. economy. U.S. manufacturers are among the most efficient and productive in the world and many of our products are unequalled in quality and price. More than 20 percent of economic growth in the last decade–the largest share –came from manufacturing.

The U.S. economy is striving to shake off recession but manufacturing lags behind. Profits have not recovered fully, exports fell steeply last year and have since seen only modest growth, and more than two million manufacturing jobs have been lost. The current recovery is the slowest of any recession since World War II. Until manufacturing is back on its feet, the overall economy will drag too.

The good news is productivity growth. Over the past four years, manufacturing averaged an astonishing 4 percent annual growth rate, more than twice that of other non-farm businesses. Productivity growth does many things, among them it:

- allows companies to increase real wages-without stoking the fires of inflation;
- demands ever new technologies and innovation, which is the differ entiator between manufacturing and other sectors;
- raises the educational level of the workforce as employees learn to use new technologies and equipment, and
- is the key to sustainable economic growth.

Innovative manufacturers who know how to implement productivity growth in their companies–by using many of the tools on display here at National Manufacturing Week–create the real wealth in the economy. This wealth helps launch new enterprises and creates jobs. Some 17 million manufacturing jobs support nine million jobs in retail, wholesale, finance, agriculture, and construction.

The National Association of Manufacturers is working to strengthen manufacturing by promoting policies to expand exports and to foster a more pro-growth environment on the home front. Our two top priorities for exports have been to help ease the overvalued dollar back down toward fair market level, and to gain U.S. participation in more trade agreements. We have been



Productivity is growing, Jasinowski says.

successful on both fronts: the dollar is coming down and Congress has enacted Trade Promotion Authority, giving President Bush the authority he needs to negotiate favorable trade agreements.

Our domestic agenda is focused primarily on tax cuts to stimulate new capital investments and encourage confidence in the economy; an energy policy to assure reliable supplies at stable prices; controlling costs associated with health care and excessive litigation, and workforce skills development, focused on drawing more talented youth into manufacturing careers.

These initiatives can help promote investment and growth. In the end, working Americans will make the difference and determine whether we will remain the economic leader of the world. National Manufacturing Week is a good place to see that we are firmly on that road.

To learn more about the NAM, contact us at (202) 637-3000, or visit us at www.nam.org.

Major sponsors for National Manufacturing Week

The following companies are the major sponsors of National Manufacturing Week at McCormick Place, which is comprised of The National Design Engineering Show, National Plant Engineering MRO and Management Show, National Industrial Automation Show, National Enterprise IT Show, Tech Transfer, and CleanTech International Cleaning Technology Exposition.

ABB
ASCO
Baldor Electric
Bosch-Rexroth
Cisco Systems
Danaher Motion
EDS
Grainger
JD Edwards
PTC
THK
Microsoft
MRO Software
National Association of Manufacturers
National Instruments
Peoplesoft
Phoenix Contact
SAP
Siebel Systems
Solidworks
Sun Microsystems
UPS

Design Engineering Show Products

(Continued from page 8)

Precision motion components

A line of high precision products includes cooling fans, miniature ball bearings, and step, spindle and DC brushless motors. AC and DC axial fans and blowers are offered 25 to 172 mm diameter, as well as hybrid and permanent magnet step motors. Applications include computing, communications, medical equipment, home appliances and automotive components.
NMB Technologies
Booth 5933
www.nmbcorp.com

Anti-backlash nuts

An anti-backlash nut option available for the company's linear motion products eliminates the small backlash between the nut assembly and leadscrew. Available for captive and non-captive designs, anti-backlash nuts are reportedly ideal for applications requiring extreme precision over long periods. Nuts are available for .218 and .250 inch screws, for the company's size 17 linear actuators.
Haydon Switch & Instrument Inc.
Booth 5140
www.hsi-inc.com

Prototyping upgrades

Upgrades for the FDM Titan™ high performance rapid prototyping system include improvement to surface finish resolution and support/removal automation for ABS modeling. The system produces polycarbonate models with layer thickness resolution of .005 in. Waterworks™ automated support/removal process uses water soluble material that dissolves after the modeling process.
Stratasys Inc.
Booth 7542 www.stratasys.com

Linear slides

Precision Ironman ball and roller linear slides offer accuracy and repeatability using friction free travel and high rigidity. Features include wedge tapered steel block preload system, factory preset, and single set-screw adjustment. The wedge provides uniform loading across the entire slide in a single step, allowing custom applications requiring tighter preloads and extended slide life.
Quality Vision International Inc.
Booth 5146 www.agslides.com

Solenoid valves

High flow Skinner 71417 and 71477 4-way solenoid valves are direct acting and designed to operate at high temperatures. Features include a high performance operator assembly, brass body, dual frequency rated coils, optional locking manual override and optional flow control. Mountable in any position, they require no minimum operating pressure to shift position and are compatible with lubricated air, inert gases, light oils and water.
Parker Hannifin
Booth 1612 www.parker.com

AmAlOx material

A line of advanced ceramic engineering materials features AmAlOx 68 alumina, a ceramic with reportedly excellent wear and corrosion resistant properties. AmAlOx material offers good strength and fracture resistance. Its range of uses includes chemical process equipment, paper, automotive, medical, aerospace, and robotics. Specific applications include pump plungers, wire guides, wear inserts, nozzles, liners,S and bearings.
Astro Met Inc.
Booth 7919 www.astromet.com

Servo motion controls

This family of servo motion controls features MintMT Multi-Tasking software, which integrates motion control, communications, networking, I/O handling, and operator interface. Operating 400% faster than other models, the motion control models range from 5 to 54A peak with 115/230-460V ac input, with common user interface from both software front end to hardware connections.
Baldor Electric Co.
Booth 5100
www.baldor.com

Linear slide

The Hiwin Linear Stage slider is actuated by a motor-driven ballscrew and guided by a linear guideway with a U-shaped rail. According to the company, the slider acts as the guideway's block and the ballscrew's nut. Reportedly easy to install and maintain, the lightweight slider offers high stiffness and high accuracy.
Hiwin Corp.
Booth 5418 www.hiwin.com

Power rocker switch

CR Series power rocker switches with 20 amp capability are available in a variety of colors, and lighted or non-lighted. Non-lighted options include single-throw and double-throw devices and a 2-color molded button. The units have an 85C maximum operating temperature and quick-connect tabs, and optional splash-resistant design. CR Series standard versions are UL, CSA, and VDE approved.
Cherry Electrical Products
Booth 6554 www.cherrycorp.com

PLC adapter

The Varioface PLC-V8L adapter is designed for 14mm relays with 6A DPDT contacts, high inrush currents and 10A SPDT load currents. The company says the adapter reduces the time needed to connect relay interfaces by 60 %, using controller system cabling. The V8L connects 8 relays through the coils without tools. A bridge plug system bridges voltage potentials and the pluggable relays are reportedly easy to replace without disconnecting wiring.
Phoenix Contact
Booth 4701
www.phoenixcon.com

(Continued onpage 12)

Design Engineering Show Products

(Continued from page 11)

Power connector

A superior pin and socket system is available with the M23 liquid tight circular power connector, the company says. With built-in strain relief, the NEMA 4X connector accommodates 7-14mm diameter cables and is available in elbow, straight or swivel snap elbow. Rated for 8-28A, 800V and 300V per pin. Integrated springs contact pins in several places for better contact.
Sealcon
Booth 5947 www.sealconusa.com

Rotary vane pump

The VACFOX high vacuum rotary vane pump is ideal for heavy duty industrial use, the company says. Available in 3 styles, the pump is low noise and easy to maintain, with a 5-year vane warranty. Applications include packaging, degassing, filling, pick-and-place, and pneumatic conveying and processing.
RIETSCHLE INC.
Booth 7511
www.rietschlepumps.com

Push-on trim seals

Push-on trim seals are reportedly the fastest and easiest way to weatherstrip with a finished look. StanPro® seals have positive grips that lock trim in place. Materials include dual durometer sponge and solid rubber with flexible wire core; dual durometer EPDM rubber with flexible segmented steel core; and vinyl and sponge rubber with segmented steel core. Edges stay solid and reportedly rattle free.
Cooper-Standard Automotive
Booth 7609
www.stanpro.com

Servo gearhead

A shaft mounted servo gearhead that is designed to save space offers an alternative to standard gearhead mounting. The right-angle design puts the servo against the machine frame and reportedly eliminates a drive shaft coupling. The output bore is available with spline, keyway or shrink disc for a zero backlash connection.
DieQua Corp.
Booth 5154
www.diequa.com

Workholding components

A positioning and operating components line is available, including bumpers and rollers, and modular workholding equipment. Solid and DuraSoft® rollers are available in various diameters and cover materials, including black neoprene, white nitrile and polyurethane. Standard mounting includes regular bearing, precision bearing, shaft drive and brake roll/clutch bearing. A variety of durometers and shapes are available.
Fairlane Products Inc.
Booth 6461
www.fairlaneprod.com

Cable carrier

E-Z Triflex® cable carriers combine 2 patented designs for flexibility and easy assembly, the company says. The carriers are ideal for robotic, space restricted and multiple-axis applications where they can snake in up to 3 directions. A crossbar design of polymer pieces allows less restricted movement of cables and hoses that are pressed into the carrier without tools. Links can be added or removed with a standard screwdriver.
igus inc.
Booth 5121
www.igus.com

Polymer filter material

High-temperature and corrosive environments demand filter materials that can resist destruction, like the PEEK™ polymer filter screen product line. Available in screen with screen sizes 35-1,000 microns, and in fabric with filter ranges from 12 microns. The products outperform polyamide and polyester, the company says. Available in rolls, narrow slit goods, tubular and stamped shapes.
Sefar America
Booth 6061
www.sefaramerica.com

Piston compressors

The 2660 series of WOB-L® piston compressors features a one-piece cast aluminum head design that reduces sound levels and eliminates leakage. Compressors report a maximum pressure of 160 psi, maximum vacuum to 29.2" Hg, and maximum free flow to 5.60 CFM. Compressors also have nine-blade fans for optimum cooling and a common head and housing platform for reportedly easy customization. Applications include medical equipment like oxygen concentrators, nebulizers, surgical equipment, and plasma separators, and such in-dustrial applications as vacuum furnaces and refrigerant recovery units.
Thomas Industries Inc.
Booth 7511
www.thomasind.com

High-speed actuator

The type SKR actuator is a high-speed product using caged technology in its guide and ball screw. CT isolates the load-carrying, recirculating elements in a cage or pocket that prevents ball-to-ball contact. Benefits include increased speed, increased accuracy, low noise and maintenance-free operation, the company reports. Providing high rigidity and little dust-generation, type SKR is available in outer rail links 150-740mm and stroke range 55-608.5mm.
THK America Inc.
Booth 5507
www.thk.com

Miniature solenoid valve

The 2-way High Flow Piloting Solenoid Valve is a miniature, low power consumption sole-noid with a different plunger design and materials in the magnetic circuit. Available in single and dual coil configurations, the valve weighs 4 oz, operating on 7.8W at 28V dc with 3,000 psi operating pressure, flowing up to 6.0 GPM. Made with the company's patented Multi-Seal®, the valve eliminates conventional O-rings and back-up rings and allows two ports to be sealed in the same axial plane, simplifying port layout.
The Lee Co.
Booth 6350
www.theleeco.com

Metric ball screws

Precision rolled metric ball screws, matched with low-profile anti-backlash nuts, are reportedly cost-effective linear motion components. Screws are available with machined or unmachined ends, to 3.0m long. Low-profile nuts are best for screw system applications is small spaces. Mounting accessories are available. Available screw sizes include 16mm and 25mm, with leads 2.5, 5, 10, and 20mm.
Techno Inc.
Booth 7402
www.techno-isel.com

CAD solutions

CADKEY Version 21 product line consists of three Windows® freeform programs. CADKEY Workshop™ Solid and Surface Modeling software is a freeform modeling solution that allows users to create, import, and edit 3D models and draw-ings, featuring a PureGeometry™ foundation, a dimensional accuracy assurance system, and a versatile assembly management and data organization structure. CADKEY Workshop EX™ has translators for Pro/E®, Ca-tia®, and Unigraphics® available as a feature, and not an add-on. CADKEY GraphX™ advanced mechanical drawing software allows users to generate drawings from solid, surface, and polygon models or create drawings from scratch.
Cadkey Corp.
Booth 7108
www.cadkey.com

Ceramic balls

CERBEC®] ceramic balls reportedly last longer and offer better performance than steel balls in bearing assemblies. Ceramic balls offer lower thermal expansion, smoother surface, increased hardness, lighter weight and corrosion resistance. Applications include linear guideways, medical and dental devices, semiconductor equipment, spindles, robotics, electric motors, power tools and ballscrews. The use of ceramic balls can lower total operating costs, the company says.
Saint-Gobain Advanced Ceramics
Booth 6331
www.cerbec.com

I/O modules Industrial Automation Show Products

Fanless computer

Panel 1060-935, from Axiom Technologies, is a fanless industrial panel computer that features a 5.7-in., QVGA, STN, LCD touchscreen and a 206-MHz Intel SA1110 StrongARM processor that is based on reduced instruction set computer (RISC) architecture. It also has 32 MB of onboard SDRAM; 16 MB of onboard Flash memory; a 24-bit digital I/O connector; and an onboard CompactFlash Type II socket.
Axiom Technology Inc.
www.axiomtek.com
Booth 4438

Flat-panel monitors

Christensen Display Products will exhibit a major extension of its of industrial flat-panel monitors. Near-Field Imaging (NFI) touchscreen technology components from 3M/Dynapro now can be integrated into most of the Christensen product family. This includes all bezel configurations and the recently released Class I Division 2-rated products. The line extension includes 28 new part numbers. Available in clear and antiglare finish, these NFI-based touchscreen monitors are mechanically identical to standard and resistive touchscreen-based products.
Christensen Display Products
www.christensendisplay.com
Booth 4212

Enclosures

Fibox reports that it manufactures the broadest line of NEMA 4X plastic enclosures in polycarbonate, ABS, and fiberglass. Many models feature UL approvals. Choose from over 15 enclosure families and more than 500 sizes to meet virtually every need. Finland-based Fibox Oy Ab manufactures its enclosures in accordance with ISO 9001 Quality Management System.
Fibox Enclosures
www.fiboxusa.com
Booth 4108

Thin-client terminals

Phoenix Contact's new line of thin-client terminals are designed for use in harsh industrial environments. Thin Clients are embedded PCs without any rotating media, which increases immunity against mechanical vibration or shock. Thin clients feature a rugged cast-aluminum housing in IP54/65 protection, high-quality color TFT displays with touchscreen option, powerful 750 MHz CPUs, built-in Ethernet interface and choice of embedded operating systems, such as Microsoft Windows CE or XP. Thin clients can be used as Web panels with integrated browsers or as remote operating stations.
Phoenix Contact
www.phoenixcon.com
Booth 4701

PDA software

GE Fanuc Automation's Cimplicity Machine Edition Logic Developer PDA software allows users to interface a Palm handheld device to any GE Fanuc Series 90 and VersaMax PLC. With Logic Developer PDA, users can perform PLC monitoring, diagnostics and maintenance quickly and easily at any PLC location in the manufacturing facility, which can save time and increase productivity.
GE Fanuc
www.gefanuc.com
Booth 4412

Plug, play, and calibrate systems

Sensotec's plug, play and calibrate systems are compliant with IEEE 1451.4 standard for connectivity of sensors and associated signal conditioning. The transducer electronic data sheet (TEDS) containing sensor specifications and calibration data is stored in the sensor. When connected, the sensor is interrogated for the TEDS information and automatically sets up and calibrates the signal conditioning.
Sensotec Inc. www.sensotec.com
Booth 4842

Ethernet switch

Moxa's EtherDevice Server ED6008 Series is an 8-port, 10/100M industrial Ethernet switch that allows connected devices to be managed and monitored via a web browser. Designed to work reliably from 0 °C to 60 °C, the switch features automatic IP recovery for device replacement, and fast spanning tree protocol to keep automation systems running continuously.
Moxa Networking Inc.
www.moxa.com Booth 4534

Short position sensor

MTS Systems Corp.'s Temposonics L Series LD4 position sensor has what is reported to be the shortest installed height available. Its short sensing element and overlapping electronics module allows mounting the complete assembly inside an actuator cylinder endcap. For compact smart cylinder applications, the LD4 position sensor has a small operating envelope for easy installation: simply fasten it to the connector. The sensor's design also allows users to install and maintain the sensor without disassembling the complete cylinder.
MTS Systems Corp. www.mtssensors.com
Booth 5348

DSP-based servo drives

PCE800 series servo drives, from Pacific Scientific, part of Danaher Motion, are DSP-based, featuring 480-V ac operation, which makes the drives powerful enough for big jobs and compatible with European voltage requirements. These features suit new installations, or a retrofit drive option in applications, such as electronic assembly, semiconductor manufacturing, robotics, packaging, and printing.
Pacific Scientific/Danaher Motion
www.danahermotion.com
Booth 5112

National Industrial Enterprise IT Show Products

Web-based ERP solution

Adonix announces two enhancements to its X3 ERP suite, an easy-to-implement, Web-based ERP solution for mid-market enterprises. The company has bolstered its warehouse management and data collection capabilities to address the needs of mid-sized companies with demanding warehouse requirements. In addition, Adonix will bring the Adonix X3 ERP software suite to the WebSphere-Express e-Business platform. The suite is said to offer rapid implementation with no hidden fees, flexibility, and strong integration.

Adonix
Booth 1017

Fixed asset accounting software

Best Software introduces its Construction-in-Progress (CIP) module-providing multiple levels of detail tracking and built-in reports, for its FAS fixed asset management solution. Users can manage and report on unlimited projects, including current status, actual versus budget variance, and project detail. With the fully integrated CIP solution, fixed assets are instantly moved to the FAS Asset Accounting system at project completion. Best Software also announces the latest release of its HR, payroll, and compliance software solution, Abra Suite 7.1. According to the company, the seamlessly integrated database, Crystal reporting, and secure query tools help businesses increase productivity and tackle HR issues.

Best Software
Booth 407

Manufacturing software solution

Epicor Software Corp. is showcasing its latest manufacturing software releases-Vista 6.0 and Vantage 6.0. Epicor's integrated manufacturing solution, Vantage, addresses the business needs of larger make-to-order, mixed-mode, and configure-to-order manufacturers by providing fully integrated relationship management applications that enable users to better manage interactions with customers and suppliers. A desktop system for small job shops or make-to-order departments of large enterprises, Epicor's Vista solution now offers fully integrated contact management, inventory management, and quality assurance capabilities.

Epicor Software Corp.
Booth 712

Business-process focused enterprise system

Exact Software offers Macola ES, the latest edition of its enterprise resources planning system solution. With this release, Exact embeds a set of business process management capabilities that reportedly allow organizations to fully integrate front- and back-office processes. By incorporating customer definable workflow templates, customer relationship management, project management, human resource management, real-time e-financials, and document management, organizations have integrated tools to better manage their business.

Exact Software
Booth 617

Three-tiered lifecycle management

IFS announces a new lifecycle management solution that reaches beyond product lifecycle management (PLM) to provide better integration and management of customers, products, and resources throughout their lifecycles. The solution, known as 3LM, is an integrated feature of IFS Applications 2003, the latest version of IFS' component-based suite of business applications. With 3LM, companies reportedly can more quickly make business decisions based on overall profitability and complete lifecycle support. The 3LM solution is said to bring together customer relationship management, product development, production, and supply chain management in one integrated package.

IFS
Booth 400

HR software

Kronos recently launched Workforce HR, allowing companies to support their strategic business objectives by streamlining human resources processes including benefits administration, personnel tracking, training tracking, health and safety, position management, organizational management, attendance tracking, and point-in-time reporting. Kronos also recently launched Workforce Payroll, enabling companies to manage payment of wages and other compensation. Kronos is a single-source provider of employee-centric solutions including integrated human resources, payroll, and labor management capabilities, providing one integrated solution that unites HR, payroll, and labor management applications, and offers comprehensive, customizable reporting.

Kronos
Booth 834

Services software package

Lilly Software Associates offers a services package called Throughput Improvement Primer (TIP). Leveraging VISUAL DBR, an integral part of Lilly Software's VISUAL enterprise software system, TIP helps VISUAL customers realize the value of implementing Theory of Constraints (TOC) and Drum-Buffer-Rope (DBR) management techniques, while minimizing the cost of their initial investment. The TIP package includes four days of TOC training, where management and production teams work together to identify their bottleneck resources, discuss how to apply the DBR methodology against these resources, and recognize the impact of these changes on their financial bottom line.

Lilly Software Associates
Booth 608

Business intelligence tools

SYSPRO announces a new business activity monitoring software tool. Business Alerts scans mission-critical information systems on a 24/7 basis to notify functional areas of the enterprise and supply chain in real-time via LAN, WAN, Internet, and wireless technology. Alert messages can be configured in HTML, Text, RTF, and XML formats, and may be sent as often as required based on pre-configured rules (i.e., continuously or daily, weekly, or monthly at specified times). Business Alerts works with a company's existing data, including Microsoft SQL, Microsoft Access, or any Microsoft/ODBC or C-ISAM compliant product.

SYSPRO
Booth 404

New CRM solution

Microsoft Business Solutions offers its new customer relationship management (CRM) solution, Microsoft CRM. The system comes in two editions: Microsoft CRM Professional Edition, which enables more complex CRM functionality including work-flow rules, customization, and back-office system integration, and Microsoft CRM Standard Edition, which provides a broad range of CRM functionality for a stand-alone CRM environment. Also featured at the show is the integration of Preactor International's constraints-based, graphical, finite-capacity scheduling application with the Microsoft Business Solutions Great Plains enterprise system.

Microsoft Business Solutions
Booth 812

Fully integrated SCP suite

Prescient, which targets consumer products companies, announces Prescient 5, one of the industry's first fully integrated supply chain planning suites. The product is developed with a common data model and user interface across all modules to ensure rapid implementation, ease of use, and low total cost of ownership. Prescient 5 includes new modules such as Dynamic Reporting and enhancements to the Advanced Planning & Scheduling application.

Prescient
Booth 730

Compact industrial printer

Intermec Technologies Corp. (booth 1034) recently introduced the EasyCoder F4 Compact Industrial printer, available in direct thermal or thermal transfer models, and runs on a 32-bit RISC processor that prints eight inches per second with fast first-label-out print speed. It is fully programmable for use with or without a personal computer. The F4 Compact Industrial can be networked with the addition of an EasyLAN 100i network interface card, which provides a high-speed multichannel interface that enables remote printer management and full bi-directional communication.

Intermec Technologies Corp.
Booth 1034

ViaWare WMS enhancements

Provia Software released four tools included with Provia's ViaWare WMS product, to streamline WMS implementations, upgrades, and ongoing testing. Data Manager simplifies warehouse configuration for implementation, and manages the data needed for subsequent sites. Gateway Simulator reduces integration efforts between the WMS and a clients' ERP or host system. Template Manager allows for language personalization and vertical industry templates for rapid configuration of the WMS. Warehouse Simulator stress tests the warehouse configuration, ensuring optimal system performance.

Provia Software
Booth 513

Supply chain execution system

SoftBrands offers DemandStream, an internal and external supply chain execution system that enables "Demand-Pull" scheduling and material flow throughout the enterprise-wide supply chain. The modular design that makes up the supply chain execution software can be implemented as a total system that will manage the entire supply chain in a Demand-Pull environment, or the individual modules can be implemented over time as part of a migration to Demand-Pull and/or Lean Manufacturing. The solution is said to reduce cycle times, inventory, and related operational expenses by improving operations.

SoftBrands
Booth 734

Plant Engineering & Facilities Management Show Products

Digital multimeters
XR-Series digital multimeters feature an ergonomic holster with a functional Magne-Grip hanging strap and are designed for electronics test, field service, and industrial applications. Five models are available.
Wavetek Meterman
metermantesttools.com
Booth 2212

Docking station
Docking station is an automated instrument management system that performs a user-prescribed regimen of function testing, full calibration, data downloading, battery charging, and record keeping functions on single and multi-gas monitors. Plant facilities and multi-location companies are able to track logged instrument readings, maintenance routines, and calibration records in a centralized database for preventive maintenance interpretation.
Industrial Scientific Corp.
indsci.com
Booth 2615

Clamps
Zinc-plated steel and stainless steel clamps are tamper resistant and have a low profile posture while maintaining a powerful constant seal, which will hold against pressure, vacuum, vibration, and fluctuations in temperature. Clamps do not damage hose/tubing materials.
Oetiker, Inc.
oetiker.com
Booth 2430

Neutralization tank
A-E series neutralization tank is a seamless one-piece polyethylene or propylene that includes a heavy treadplate and integral threaded manway for easy removal and inspection.
Schier Products Co.
schierproducts.com
Booth 2610

Condition monitoring
Ultrawave 170 can monitor acoustic vibration from bearings as well as contact and non-contact infrared temperature, RPM, audible noise, and a host of other parameters. The all-in-one predictive maintenance device offers versatility for today and expansion possibilities for tomorrow.
SDT North America
sdtnorthamerica.com
Booth 1544

Process instrumentation
The 789 ProcessMeter combines a digital multimeter and a loop calibrator in a single handheld tool. Features include a HART mode setting with a built-in 250-ohm resistor, increased battery performance with four AA batteries, 0-100% mA span check buttons to toggle from 4 to 20 mA, and an infrared I/O serial port compatible with FlukeView charting and analysis software.
Fluke Corp.
fluke.com
Booth 2219

Relay connectivity
Varioface PLC-V8L adapter is designed for the 14-mm relay with 6-A DPDT contacts, high inrush currents, and 10-A SPDT load currents. The adapter makes connecting relay interfaces to PLC and DCS output cards easy and quick using Controller System Cabling.
Phoenix Contact Inc.
phoenixcon.com
Booth 4701

Filters
Finite Filter's H-Series is the right solution for most compressed air/gas applications and can greatly improve systems. Housings are available with oil removal, particulate, and oil vapor removal elements.
Parker Hannifin
parker.com
Booth 2507

Vibration analysis
National Remote Diagnostics is an internet-based tool that creates standardized reports to identify where problems reside by piece of equipment so maintenance professionals can immediately take the corrective steps to resolve the issues without worrying about the tedious analysis process. Vibration analysis information is available 24 hr a day, 7 days a week.
National Electrical Carbon
vbseries.com
Booth 1429

Pipe repair
STOP IT pipe repair system has been used to repair leaks in chemical plants, refineries, processing plants, manufacturing plants, municipal water and wastewater utilities, electric utilities, and in other applications. For operations where piping carries flammable, toxic, or corrosive products, the system provides a method to restore product flow in minutes and could stop an environmental contamination problem before it starts.
Indumar Products, Inc.
stopit.com
Booth 2314

Access solutions
Lapeyre Stair manufactures alternating tread stairs that reduce risk of injury, improve labor productivity, and conserve expensive floor space. Product lines include original fixed stair, mobile stairs, crossover systems, and exit and intermediate platforms.
Lapeyre Stair, Inc.
lapeyrestair.com
Booth 2542

(Continued on page 16)

Plant Engineering & Facilities Management Show Products

(Continued from page 15)

Safety device
I-Zone area detection system uses infrared technology to sense pedestrians before they are in the door opening and prevents a door from beginning to close when a person enters the 7-ft safety zone. Each IR sensor looks out in front of both sides of the opening, and the infrared beams cross to form the safety zone.
Rite-Hite Doors, Inc.
rite-hite.com
Booth 1605

Infrared camera
The ultra-compact ThermaCAM E2 is a smart infrared camera with real-time thermal imaging, temperature measurement, and JPEG image storage. The product is affordable and weighs less than 1.5 lb.
FLIR Systems, Inc.
flirthermography.com
Booth 2319

Chain hoists
A line of single-phase electric chain hoists and motorized trolleys are recommended for temporary job sites, machine shops, and maintenance areas where three-phase power is not available. SNER hoists and SMR trolleys provide a heavy-duty rating for tough jobs, a lightweight and low headroom design, easy maintenance, and capacities through 3 tons.
Harrington Hoists, Inc.
harringtonhoists.com
Booth 1642

Surge suppressor
Surge3 is a solution for overvoltage protection of industrial electronics, allowing for simple yet effective installation to protect PLC, HMI/VCP (video control panel), PID controllers, industrial computers, variable speed drives, and vision systems. MOV and Diode are two technologies offered, both of which are interchangeable with a plug-in suppressor and common base mount.
Cooper Bussmann
bussmann.com
Booth 1820

Cooling unit
Port-A-Cool HP model portable personal cooling unit is constructed of rust-proof, leakproof rotomolded plastic and employs evaporative cooling to produce cool air instead of circulating hot air. The HP model produces 10,500 cfm and offers variable speeds.
General Shelters of Texas
generalshelters.com
Booth 2416

Retrieval system
Hi-speed Router storage and retrieval system stores and retrieves palletized aluminum sheet stock and interfaces with a high-speed router. Pallets are automatically delivered to the router as needed, allowing for quick changeover and inventory management of materials.
Webb-Stiles Co.
webb-stiles.com
Booth 1834

Maintenance consulting
Life Cycle Engineering is a maintenance engineering consulting firm that has developed hundreds of planned maintenance/equipment reliability programs. Benefits of maintenance solutions have included application of best practices, improved maintenance planning and scheduling, enhanced productivity and quality, reduced capital expenditures, and decreased equipment downtime.
Life Cycle Engineering
lce.com
Booth 1937

Safety mat
An ergonomic presence-sensing safety mat is designed to reduce worker fatigue while guarding personnel against injury from hazardous machinery. Comfort Step mat enables workers to use the mat as a traditional safeguarding device, stopping dangerous machinery to change tools or parts, and is ideal for normal foot traffic in tough industrial environments, including ones where fluids may be present.
Larco
larcomfg.com
Booth 2137

Gaskets
TriGuard gaskets are suited for positive sealing of aggressive fluid piping applications on fragile flanges or flanges with irregularities, where bolt torque and gasket compression stress must be limited. Gaskets are used for contamination and highly aggressive fluids, including chlorine.
W.L. Gore & Associates, Inc.
gore.com/sealants
Booth 2517

Air dryers
Balston SMART Dryer membrane air dryers offer lower operating costs and better performance than both noncycling and cycling refrigerant air dryers and eliminate downtime and costly repairs resulting from dirty, wet compressed air supplies. Products provide dry compressed air at a guaranteed dewpoint of 35 F and flow rates up to 200 scfm.
Parker Filtration
balston.com
Booth 2407

Wastewater evaporators
Wastesaver and SWE-II Sentry System can efficiently and economically reduce complex wastewater chemistries, using an integrated PLC to control the process and maximizing volume reduction typically costing 2-6 cents/gal. Systems offer redundant safety sensors, three-stage mist elimination, level control, and options for auto drain, auto decant, and remote modem monitoring.
Severn Trent Services-Samsco
Booth 2419

Electrical testing
Model 61-704 combines a 200-A clamp/multimeter, noncontact voltage tester, voltage tester with shaker, and harmonics tester in one easy-to-use tool. The versatile ergonomic clamp-style tester enables electricians to perform testing procedures that previously required four separate tools, resulting in improved cost efficiencies, a less cluttered tool bag, and enhanced on-the-job safety due to the tester meeting demanding CAT III standards.
Ideal Industries, Inc.
idealindustries.com
Booth 2024

Nozzles
PNR America offers a complete line of industrial spray nozzles and accessories, including full, flat, spiral, and hollow cone hydraulic nozzles, air atomizers, tank wash, eductors, and spray dry.
PNR America
pnramerica.com
Booth 2428

Shaft alignment
smartALIGN brings simplicity to shaft/coupling alignment by using smart navigation with a joystick and two keys, which eliminates the need for operators to memorize a sequence of keystroking. Onscreen help text guides the operator to the results for best corrective shimming and horizontal moves.
Ludeca
ludeca.com
Booth 2432

Containment Station
Containment station Gator iDAC is an integrated dispensing and containment station for IBCs that captures and spots spills or leaks in its exclusive open-view sump. Unique design offers a 400-gal sump capacity and chemically resistant, Gator-Tuff polyethylene construction for a load bearing capacity of 8,000 lb.
Justrite Manufacturing Co.
justritemfg.com
Booth 2728

Industrial control
A full line of radio controls ranging from 4, 6, and 8-button Protean control to the dual joystick J series radio for five-speed, eight-motion applications are available. Wireless technology incorporates the latest advancements in RF and microprocessor design and includes features such as watchdog error circuitry, output relay monitoring, and programmable password protection.
Insul-8 Corp.
insul-8.com
Booth 1636

Cooling towers
OBR Cooling Towers is a contractor specializing in cooling tower refurbishment and repair. The company services all makes and models of cooling towers and sells parts for all towers.
OBR Cooling Towers, Inc.
obrcoolingtowers.com
Booth 2544

Storage cabinets
ID Casework is a laminate-clad, industrial storage cabinet. These extra-deep cabinets are available in several configurations to meet storage needs.
Stevens Industries
Booth 2242

Metal distributor
Metal Supermarkets provides small quantities of all types of metal, cut to size, including hot roll, cold rolled, aluminum, stainless, and brass. Same-day pickup or fast delivery are available.
Metal Supermarkets
metalsupermarkets.com
Booth 2434

Equipment maintenance
Alfa Laval features a comprehensive program of maintenance, repair, and spare parts for heat exchangers, separators, and sanitary flow equipment, including installation and startup, onsite troubleshooting and repair, maintenance agreements, parts management, and repair/exchange for major components.
Alfa Laval
alfalaval.com
Booth 2427

Insulation tester
MEG10-01 is a useful addition to any maintenance regime and includes five selectable test voltages up to 10 kV, exclusive combination digital/analog display, storage and downloading, an integral timer, and automatic Polarization Index (PI) test. The tester is operated from either line or rechargeable battery power.
Megger
megger.com
Booth 2529

Industrial guide
Ameritech Industrial Purchasing guide is an industry-focused publication distributed to industrial purchasers, specifiers, and key decision makers.
SBC Ameritech
Booth 2036

Status indicator
Unistat LED Status Indicator is compact, attractive, and easy to install and offers four colors and four flash patterns from a single light source. Units are available in 24-Vac/dc and 120-Vac operating voltages.
Federal Signal Corp.
federalsignal-indust.com
Booth 2330

Maintenance software
Technicians can perform a multitude of tasks with work order systems utilizing a palm handheld computer. The maintenance department becomes mobile and paperless.
TMA Systems
tmasystems.com
Booth 1327

Flow-through storage
Flow Cell is a modular rack and track unit used for flow-through storage for material handling workstations. It is ideal for assembly line picking and work-in-progress applications.
Unex Manufacturing, Inc.
unex.com
Booth 2039

Doors
Nergeco USA manufactures custom-made productivity doors specifically designed for a wide range of industries. Doors offer efficient climate control and reduced energy consumption, a soft-bottom edge for maximum safety, a tight seal to resist wind and insulate from surrounding environments, and vision panels capable of providing full-width visibility.
Nergeco USA
nergecousa.com
Booth 2209

Smoking shelter
Modular, fully enclosed Air-Flo shelters suit any site and economically accommodate staff and visitors year-round to eliminate concerns of nonsmokers, smokers, and management. Options include lighting, heating, air-conditioning, exhaust systems, benches, ash urns, and insulated panels for sun control.
Duo-Gard Industries, Inc.
duo-gard.com
Booth 2241

Screwdriver blades
slipSTOP blades prevent the screwdriver from slipping out of the fastener's recess thanks to a laser-cut pattern on the drive tip that creates a rough surface to bite into the screw recess. Chances of operator injuries are considerably reduced.
Wera Tools, Inc.
weratools.com
Booth 2239

Lubrication
Lubrication tools bolster machine reliability by keeping valuable lubricants free from dirt, dust, sand, water, and other machine-damaging contaminants. The unique sealing system also works in reverse, preventing dangerous and costly oil spills.
Oil Safe Systems, Companion Products, Inc.
oilsafe.net
Booth 2614

SHOW PRODUCTS

Leak detector
Blue Max Plus is a handy fluorescent leak detector that clips onto a shirt pocket. It features an adjustable service mirror and a unique telescopic extender, providing a reach of 28 in. This makes it easier to pinpoint hidden leaks in hard-to-reach parts of air conditioning and refrigeration units that are off limits to larger lamps.
Spectronics Corp.
spectroline.com
Booth 2027

Maintenance management
Caver-Morehead Systems, Inc., is a computerized maintenance management system software developer and engineering firm that sells and services its software programs to contractual clients.
Caver-Morehead Systems, Inc.
caver-morehead.com
Booth 1248

Nitrogen generator
MAXIGAS nitrogen generator offers an innovative solution to nitrogen gas supply. A reliable source of nitrogen can be produced from existing compressed air supplies, eliminating the need for high-pressure cylinders or bulk cryogenic liquefied gas.
domnick hunter
domnickhunter.com
Booth 2327

Infrared thermometry
PhotoTemp MX6 is an integrated infrared thermometer and digital camera allowing for non-contact temperature measurements while documenting with digital photographs. Temperature, locations, and date/time are superimposed on the images when viewed on a computer or in reports.
Raytek Corp. raytek.com
Booth 2332

Surplus site
plcCenter.com buys and sells new and used surplus products to other electro-mechanical and industrial control companies to provide an affordable method for productivity elevation. Companies can save 50-80% purchasing surplus items.
Speck Industrial Controls, Inc.
plcCenter.com
Booth 2037

Tool storage
TRX4100SS 8-drawer chest and 12-drawer cabinet combination features stainless steel unibody shells, full-extension ball-bearing slides for complete tool access, and an angled Traxx cover lid design with embossed chest stiffener for increased strength. Drawer liners to hold and protect tools come standard.
Waterloo Industries, Inc.
waterlooindustries.com
Booth 2227

Wood/Metal Tool
DD 500 Tradesman is designed for craftsmen who work in wood and metal and features a rugged machine with a replaceable diamond sharpening wheel.
The Drill Doctor,
Professional Tool Manufacturing
drilldr.com
Booth 2836

Coupling
T&B XJG Rigid Conduit Expansion coupling is an innovative coupling that simplifies and speeds the job of installing expansion fittings in long runs of conduit. No disassembly is required during installation. As a result, there are no parts to drop or lose, and fewer tools are required.
Thomas & Betts Corp.
tnb.com.
Booth 1422

Face shield
ClearVisor with Adflo system combines full eye and face protection with a slim, lightweight, powered air-purifying respirator. The face shield has exceptional clarity and a wide viewing area, while providing workers protection from high-impact hazards. The system can be used in a variety of industrial settings, including furniture manufacturing.
Hornell, Inc.
hornell.com Booth 2514

Roll-out storage system
The Lista Shallow Depth Storage Wall® is a modular drawer, shelf, and roll-out tray storage system and is available in two models, standard (B251) and shallow depth (B255). A variety of different drawer heights provide storage for a range of items, from very small parts to large tools. With 100% full extension, the drawers and roll-out trays allow easy access to stored items. Both the drawers and trays can hold up to 440 pounds each. Wide span beams, designed for large bulk items, hold up to 1500 pounds and are available with a variety of deckings, including steel, wire, and plywood. The Storage Wall® can be designed to include an order/issue counter and a 12" deep, recessed bookcase for convenient storage of reference books.
Lista International Corporation
www.listaintl.com
Booth 1520

Screen door system
Bug Blocker overhead screen door system addresses security, ventilation, and pest control issues.
Rasco Industries, Inc.
bugblocker.com
Booth 2408

SHOW NEWS

Great Expectations for National Manufacturing Week 2003

(Continued from page 10)

sourcing, and Integration Professional Services: Industrial Automation Vendors Move Far Beyond Hardware," [at 9-10 a.m.] on March 4.

Stuttard adds that NMW 2003 will also have a new layout, which will sandwich the Motion Technology Area's exhibitors between NIAS and the National Design Engineering Show (NDES). This area offers a comprehensive collection of pneumatic, electromechanical, motors, power transmission, linear motion components and hydraulic components, and control systems.

New shows, pavilions

Besides its traditional four shows– NIAS, NDES, National Plant Engineering and Facilities Management show, and National Enterprise IT (NEIT) show–NMW 2003 will host two more shows. The first will be the Technology Transfer Conference and Expo, which will be a forum for buyers and sellers of technology licensing. The second will be the co-located CleanTech International Cleaning Technology Exposition.

NMW 2003 will also include several special pavilions, such as Customer Relationship Management (CRM), Fluid Handling, Contract Manufacturing, Material Handling, and Large Equipment. Located at the NEIT show, CRM will gather products and solutions for manufacturing and IT executives seeking to improve supplier and customer relations, while cutting costs, generating revenue, and retaining customers.

In addition, a new E-learning pavilion at NEIT will showcase education, e-learning, and training for manufacturers, and will also include exhibits, educational forums, and keynote presentations. Finally, a new International pavilion at NDES will include numerous international manufacturers, suppliers, and vendors.



Keynotes and conference

This year's expanded technical conference will be headlined by Frederick Gregory, NASA's deputy administrator, who will deliver NMW 2003's keynote address, "Bringing Space Technology into the Commercial Sector," at 1:15-2 p.m., March 5, in Room S101.

Gregory will outline the space agency's vision and mission objectives, its plans for future space exploration, and its efforts to transfer cutting-edge technologies to the manufacturing sector.

Besides its automation industry address, the NIAS-based conference track will include another panel, "Dick Morley and the Technology of Business: Asking the Tough Questions," at 2-3:15 p.m. on March 4 in Room S101. Morley's panel of insiders will probe the operations, technology, and ethics of today's enterprise.

Some other notable conference sessions will include: "Lean Manufacturing and Mass Customization," 10:15-11;30 a.m., March 3, by Richard Lebovitz, Factory Logic Software; and "Network Convergence in Manufacturing," a free session at 1-2:15 p.m., March 3, by Kathy Hill, Cisco Systems' vice president and Desktop Switching business general manager, who will chart Ethernet's move onto the factory floor.

Design Show 2003

3 Dimensional Engineering 7000
3Dlabs, Inc. 6501
3M 9060
3M Adhesives Technology 7500
80/20, Inc. 8028
A & A Manufacturing Company, Inc. 7439
A & E Plastics/TCI Vacuum Forming 7437
A.B. Controls and Technology, Inc. 6518
Abbott Interfast Corporation 7318
Acme-Monaco Corporation 6031
ACS Industries, Inc. 5568
ADDA USA 7937
Adron EDM Specialists 7846
Advanced Forming Technology/PCC 7536
Advanced Materials Technologies 7453
Advent Tool & Mold 7935
Air System Products 6459
Airpax 5046
AK Coatings, Inc. 5959
Alberta Economic Development 7728
Alloy Bellows & Precision Welding 5960
Alpha Gear Drives, Inc. 5342
Amalga Composites, Inc. 7338
American Excelsior Company 5868
American Metric Corporation 5151
American Ring & Tool Company 7412
Amerigo Tech. Ltd. 5246
Ametek US Gauge 7307
AMK Drives and Controls, Inc. 6137
Amorim Industrial Solutions 8039
Amveco Magnetics, Inc. 5851
Anderson Tool & Engineering Co. 5050
Antou Resource Inc. 5047
APEM Components, Inc. 5861
Apollo Enterprises, Inc. 5465
Apple Rubber Products, Inc. 7622
Area Development 7528
ARI Industries, Inc. 5759
Armoloy of Illinois, Inc. 6450
ARRK Product Development Group 8107
ASCO 5109
ASME International and Mechanical Engineering Magazine 5519
Astro Plastics 7734
Astro-Met, Inc. 7919
Atlas Engineering 5552
Autodesk 5901
Axiom Technology Inc. 8121
Axiomatic Design Software, Inc. 6307
Baco Controls, Inc. 7653
Bal Seal Engineering 7517
Baldor Electric Co. 5100
Ballscrews & Actuators 5112
BaoDing YiTong Machinery Mftg. Co., Ltd. . . . 5436
Bastech, Inc. 7118
Batesville Products, Inc. 7736
BD Accu-Glass 7614
Bead Industries 7908
BEI Technologies, Inc. 5345
Beijing Metals & Minerals Import & Export Corp. .5435
Beijing MMC Metals & Minnerals I/E Co., Ltd. .5335
Bellcomb Technologies, Inc. 5847
BIAM / General Titanium, Inc. 7808
Bimba Manufacturing Company 5136
Bird Precision 7747
Bison Gear and Engineering Corporation . . 7526
Block USA L.P. 6339
BlockMaster Electronics, Inc. 8013
Blue Chip Engineered Products 5721
Bodine Electric Company 5330
Bollhoff Rivnut®, Inc. 5931
Bosch Rexroth Corporation 5107
Bowers Manufacturing Co. 8011
Brecoflex Co., L.L.C. 5747
Brother International Corporation 7838
Buehler Motor 5758
Bunzl Extrusion - Chicago 7303
C & J Metal Products, Inc. 6062
Cable Manufacturing & Assembly Co., Inc. .9059
Cables To Go 5746
CADKEY Corporation 7108
Canadian Thermostats & Control Devices Ltd. 5553
Canfield Connector/Spartan Scientific 7648
Caplugs 7509
Cardinal Aluminum Co. 8102
Catic Guangzhou, Industrial Department . . 7934
Celesco Transducer Products, Inc. 6557
Ceradyne, Inc. 8023
CeramTec North America, Inc. 7845
CGI 7717
Changda Metal Manufacturing Co., Ltd. . . . 8132
Cherry Electrical Products 6554
Chicago Decal Company 5733
China Custom Manufacturing LTD 5564
China National Machinery & Equipment Imp. & Exp. Corporation 5335
China National Machinery Import & Export Corporation 5337
China Ningbo Huatai Electric Motor Co., Ltd. .5252
Cimmetry Systems, Inc. (CSI) 6501
CITRA 7628
Clancey Hardware & Hinge 7701
Cliff Impact Division 5732
Colder Products Company 5535
Colson Caster Corporation 7321
Colson Plastics Group 7514
Comsol, Inc. 6905
Comtec 8031
Conductive Technologies, Inc. 5445
Cool Polymers 5668
Cooper-Standard Automotive 7609
CoorsTek 8127
Creco Corp. 5721
CSB Group Holdings Limited 5435
Custom Connector Corporation 7801
Custom Products Corp. 7208
Cybershield, Inc. 7638
Da/Pro Rubber, Inc. 8003
Data Display Products 7821
Davies Molding, L.L.C. 6037
D-Cubed Ltd. 6907
Delta Technologies 6259
Deltran PT 5112
Del-Tron Precision, Inc. 5554
Derby Cellular Products 7833
Design News 5126
Designfax 6814
Desktop Engineering 6714
DE-STA-CO Industries 5928
Devon Precision Inds., Inc. 7634
Dielectrics 6818
DieQua Corporation 5154
Divelbiss Corporation 7515
Diversified Systems, Inc. 5957
Douglas Corporation 6152
Dresser Instruments 7418
Driv-Lok, Inc. 7705
Du-Co Ceramics Co. 8020
Dudek & Bock Spring Manufacturing Company 5734
Duff-Norton 5526
DuPont Teflon® Industrial Coatings 7703
Dynacast, Inc. 7531
E. N. Murray Co., Inc. 5466
E-A-R Specialty Composites 6521
East West Manufacturing 7552
Easylift of North America 5754
ebm Industries, Inc. 6131
ECN 5044
EDS PLM Solutions 5501
EGC Enterprises, Inc. 6028
EITD 5044
elastometall Kentucky LLC 8136
Electro Assemblies Corporation 7650
ElectroForce Division 5254
Electrolizing, Inc. 5538
Elesa USA Corporation 5922
Elliott Manufacturing Company, L.L.C. . . . 6253
Emhart Teknologies 5513
EMKA Incorporated 6250
Empire Level Manufacturing Corporation . . 7202
Engineered Sinterings & Plastics, Inc. . . . 7534
Engineering Industries Association 6617
Enterprise Florida 7914
EnterpriseSoftware/HQ 5511
Enthone, Inc. 7539
E-T-A Circuit Breakers 6345
Fairlane Products, Inc. 6461
Famic Technologies Inc. 6406
Farmington Engineering, Inc. 7421
Fastbolt Corporation 7600
Feller U.S. Corporation 5963
Felton Brush, Inc. 5630
Filtertek Inc 7604
Firestone Industrial Products Company . . . 7713
Flash Cut CNC 6902
Flexan Corporation 7405
Flexicon Ltd. 6618
Fluent Inc. 7103
Footmaster 8018
Force Control Industries Inc. 6531
Foremost Plastic Prod. 7615
Fuji Prescale 5652
Fulton Bellows & Components 6362
Fundigex - Castings' Exporters Association of Spain .8031
Gao Ming Tie Enterprise Co., Ltd. 8046
Gasser & Sons, Inc. 5940
Gemini, Incorporated 7452
General Extrusions, Inc. 5935
General Magnaplate Corp. 7805
Gibbs & Associates 6501
GKN Sinter Metals 7932
Globe Motors 5542
GM Nameplate 6551
Gold Shield of Indiana, Inc. 6548
Gold Star PVD Coatings 7435
Gordon Brush Mfg. Co., Inc. 7502
Gorman-Rupp Industries 7354
Grimm Brothers Plastics Corporation 6908
Griswold Rubber Company, Inc. 5838
Groov-Pin Corporation 5534
Groschopp, Inc. 5257
Gupta Permold Corporation 8111
Gurley Precision Instruments 5533
Hammond Enclosures 8017
HangZhou Metals Minerals Machinery and Chemicals Imp. & Exp. Co., Ltd. 5437
Hanser Gardner Publications 6262
Harman Corporation 8128
Harmony Castings 5853
Harris InfoSource 7205
Hartwell-Dzus Inc. 7706
Haydon Switch & Instrument, Inc. 5140
Heany Industries, Inc. 7820
Heatron, Inc. 5545
Heidenhain Corporation 6326
Hewlett-Packard 6711
Hi-Tech Controls, Inc. 5947
Hi-Tech Rubber, Inc. 7739
Hiwin Corporation 5418
Ho Tai Steel Mold Co., Ltd. 7944
Hoesch Hohenlimburg GmbH 6123
Hoover Precision Products, Inc. 5463
HYTEC, Inc. 6906
IBM 5917
IDC 5112
Igus, Inc. 5121
IKO International, Inc. 5461
ILCO Unican 6247
Illinois Lock Company 5836
ImpactXoft USA 6511
Imperial Electric 6260
Industrial Tectonics, Inc. 5516
Innovative Components 5659
Intech Corporation 5055
Integra Enclosures 5450
Intelligent Motion Systems, Inc. 7326
International Extrusions 6363
International Hardcoat Inc. 8035
International Sensor Systems, Inc. 7432
Intricate Grinding & Machine Specialties . . 7633
Iowa Laser Technology, Inc. 8037
ITW Spiroid® 5833
J & L Plastic Molding 5768
J.W. Winco, Inc. 7351
Jentech Precision Industrial Co., Ltd. 8047
Jiangsu SuSun Group Corp., Ltd. 5339
Job Shop Synergy 7519
Joyner Die Casting & Plating Company, Inc. 7636
KabelSchlepp America, Inc. 5416
Kastalon, Inc. 7302
KB Electronics, Inc. 7403
KBC Bearings Company 5152
KDL Precision Molding Corporation 8126
Key Bellevilles, Inc. 6553
Kinetics, Inc. 7549
King Wire, Inc. 7522
KIP America 6319
Klauber Machine and Gear Co. 5542
KMS Bearings, Inc. 5148
KTR Corporation 7602
Lane Techical Sales 8145
Laserage Technology Corporation 7533
Lee Spring Company 7313
Lemo USA, Inc. 7608
Limtech, Inc. 6051
LIN Engineering 5138
Linak U.S. Inc. 5460
Linemaster Switch Corporation 6523
Lomont Molding, Inc. 6900
Long-Lok Fasteners Corporation 7605
Lubrication Technology, Inc. 7651
Ludlow Composites Corporation 5857
MacDermid Incorporated 5858
Machine Design 6134
MACtac 6347
Magnetic Inspection Laboratory, Inc. 7233
MANA- Manufacturers' Agents National Association .5150
Marathon Special Products 5850
Marking Specialists Group 7745
Master Magnetics, Inc. 5739
Mastercam / CNC Software, Inc. 6514
Masterflux 7722
Maxon Precision Motors, Inc. 6161
Mayr Corporation 5459
MBC Composite Bearings Mfg., Inc. 5451
Mechanical Engineering 5519
Mercury Plastics, Inc. 7802
MET Plastics, Inc. 7720
Metal Impact Corporation 5757
Metal Powder Industries Federation 8112
Metallurgica Calvi Spa 5118
Metric & Multistandard Components Corp. .6252
MetroMark 6453
Mexico Trade Commission 7116
Meyer Retaining Ring Copmany, Inc. 7918
Michigan Steel Spring Company 6147
Micro Stamping Corporation 8009
Micro Surface Corporation 7414
MicroMo Electronics, Inc. 6045
Micron 5112
Mideal Technology Co., Ltd. 5561
Midwest Metal Products 7839
Midwest Wire Products L.L.C. 7903
Minco Products, Inc. 7619
Mini Gears, Ltd. 5250
Minnesota Rubber and QMR Plastics 7911
Minolta Corporation 7545
Minor Rubber Company, Inc. 7422
Moffatt Products, Inc. 7411
Molded Fiber Glass Tray Company 7504
Monroe Engineering Products, Inc. 6527
Mouser Electronics 7842
MTS Systems Corporation 5348
MW Industries, Inc. 7200
NAM Site Selection Network 6126
nash_elmo Industries 5735
Nason 7436
NB Corporation of America 6342
NCEMT- Concurrent Technologies Corp. . . . 7506
New Hermes, Inc. 6248
NewAge Industries, Inc. 5653
Newark Electronics 9065
Nickelson Plastics Inc. 7244
Nidec-Copal 5542
Nielsen/Sessions 7917
Nimet Industries, Inc. 8005
NMB Technologies Corporation 5933
Norgren - KIP Fluid Controls 5340
Northeast Tennessee Valley Regional Industrial Development Association 5765
Northrop Grumman Poly-Scientific 6159
Northwest Etch Technology 7818
Northwestern Tools, Inc. 5537
Noshok, Inc. 6354
Noveon, Inc. 9056
NSK Precision America, Inc. 5926

Design Show 2003

NuSil Technology 7652
Nyden Corporation 6059
Nye Lubricants, Inc. 5845
Nylok Corporation 7548
Objet Geometries, Ltd. 7445
Oconomowoc Manufacturing Corp. 5560
Ohio Development Association 7102
Optical Gaging Products, Inc. 5146
Oregon Economic Development 5952
Oriental Motor USA Corporation 6334
Orttech, Inc. 5144
Osai - USA, LLC 6223
Osborn International 5457
Owens Corning 7433
Pacific Bearing Company 5248
Pacific Scientific 5112
Packer Technologies International, Inc. 6919
Pad Print Machinery of Vermont 6145
Panel Components Corporation 7430
Paneloc Corporation 7909
Peer Bearing Company 5468
PEM Fastening Systems 5550
Petersen Precision 5753
Phillips Plastics Corporation 5713
Piab Vacuum Products 5844
PIC Design 6526
Pisco USA, Inc 5042
Pittman 5448
Pivot Point, Inc. 7317
Plainfield Companies 7708
Plitron Manufacturing, Inc. 7306
Poco Graphite, Inc. 7922
Polymer Sealing Solutions 6148
Polymer Technologies 7639
Portescap 5112
Power Electric 5950
PPS, Inc. 6050
Precico Group 7817
Precipart Corporation 7733
Precision Industries Corporation 5860
Precision Prototype and Manufacturing, Inc. 5721
Pres-On Tape & Gasket Corporation 5839
Principal Manufacturing Corporation 5721
Product Design and Development 5945
Product News Network 5511
Product Safety Consulting Inc. 7236
Profile Plastics Corporation 7100
Protex Fasteners, Ltd. 6065
PSC Computer Products 6337
PTC 5911
Qontrol Devices, Inc. 5961
Qssolutions Inc. 7006
Quality Transmission Components 8115
Queretaro State Government 7111
R.T. Enterprises, Inc. 5661
Rahco Rubber, Inc. 5636
RapidPro Manufacturing 7646
Rauschert Technical Ceramics 7920
RB HybroDynamics, Inc. 5251
RB Royal Industries 7606
Real Seal Company, Inc. 7429
Rectus-TEMA Corporation 7331
Refractron Technologies Corporation 7921
Reid Tool Supply Company 7311
Reliance Gear Co. Ltd. 7204
Respect Her Industrial Company, Ltd. 7945
RG Consulting, Inc. 6047
RIA CONNECT, Inc. 5634
Richco, Inc. 7300
Richway Industries 5736
Right Hemisphere 6903
Robinson Rubber Products Company, Inc. . 6032
Rockford Ballscrew Company 5453
Rockford Silkscreen 7916
Rogan Corporation 7447
Rogers Corporation 9057
Roland DGA Corporation 6501
Rolco, Inc. 7523
Rollon Corp. 5116
Rolta International, Inc. 7105
Ron-Vik, Inc. 8038
Rose & Bopla Enclosures 7553
Rotary Systems 5646
Rotonics Manufacturing, Inc. 7737
Rotor Clip Company, Inc. 5745
Rubbercraft 7939
Rubberlite, Inc. 8006
Rubis Precis 7829
Ruger Investment Casting 8122
S & S Hinge Company 7336
S & W Manufacturing Co., Inc. 7206
S.S. White Technologies, Inc. 5539
Sadev Decolletage 5662
Saia-Burgess USA Incorporated, Inc. 8104
Saint-Gobain Advanced Ceramics 6331
Saint-Gobain Performance Plastics Corp. . . 5530
Sanyo Denki America 6139
Sarnatech BNL 6516
SCANCON 5350
Scicon Technologies 7644
Scotland-West 6359
Sealcon L.L.C. 5947
Sealeze 6034
Sefar America, Inc. 6061
Seitz Corporation 7538
Sensor Products, Inc. 6033
Seoul Semiconductor 5157
Serigraph, Inc. 5819
Servometer - PMG, L.L.C. 5854
ShangDong Machinery & Equipment Imp. & Exp. Group Corp. 5336
Sherex Industries, Ltd. 7450
Shurflo Pump Mfg. Co. 7415
Siderval SpA 5752
SIFCO Selective Plating 7823
Signicast Corporation 7901
Siko Products, Inc. 5458
Simo Import & Export Co., Ltd. 5438
Simrit 6153
Skydex Technologies, Inc. 7544
Small Precision Tools Inc. 8021
Smalley Steel Ring Company 7408
Smart Products, Inc. 8147
Smith Metal Products 8105
Solid State, Inc. 7738
SolidWorks Corporation 6501
Solus Industrial Innovations 6452
Solvay Advanced Polymers L.L.C. 6321
Sorbothane, Inc. 7417
Southco Inc. 7701
Spencer Industries Incorporated 8001
Spicer Corporation 7005
Spirolox 5516
Spraying Systems Co. 6142
Sprinter Marking, Inc. 5546
Spyraflo, Inc. 5848
SSI Technolgies, Inc. 5742
Stabilus 5953
Standard Iron & Wire Works 7840
Standard Register/Label Solutions 7342
Stankraft, Inc. 5936
Sterling Instrument 7400
Stik-II Products 8114
Stock Gears, Inc. 7339
Stocker Hinge Manufacturing Company . . . 7537
Strain Measurement Devices, Inc. 7203
Stratasys, Inc. 7542
Sub Source, Inc. 6460
Subcontracting Exchange of Bilbao Chambers of Commerce 8031
Suhner Manufacturing, Inc. 6117
Superior Technical Ceramics Corp. 7913
Surface Finishes Company, Inc. 7936
Synmark-Vikeland, L.L.C. 5542
Syntex Rubber Corporation 6039
T.E.A. Machine Components, Inc. 5842
T.F.T. Tools, Inc. 6060
Tanis Incorporated 5536
Tapemark 8142
Techmetals, Inc. 7207
Techno-Isel Systems 7402
TE-CO 5751
Tecton Products LLC 8014
Ted Dasher, Inc. 7002
TEI Fans 6353
Teknicote, Inc. 8144
Tekscan, Inc. 7018
Telco International Corp. 5542
Tente Casters, Inc. 7337
Tescom Corporation 7314
Texas Economic Development 5528
Textron Fastening Systems 6357
Textron Power Transmission 5142
The American Ceramic Society 8022
The Fuller Brush Company 5962
The Goodyear Tire & Rubber Company . . . 5942
The Hall Company 5830
The High Ground of Texas 7104
The Lee Company 6350
The Protomold Company, Inc. 6501
The Vollrath Company 7637
Thermik Corporation 7242
Thermion Systems International 6561
Thermtrol Corp. 8008
ThinGap Motor Technologies 6222
THK America, Inc. 5507
Thomas Industries, Inc. 7511
Thomas Publishing Company 5511
Thomas Register of American Manufacturers 5511
Thomson Industries, Inc. 5112
Ticona - Engineering Polymers 6328
Timco Rubber Products, Inc. 6352
Tinnerman Palnut 7328
Tiodize Co., Inc. 6063
Tool-Less Plastic Technologies, LLC. 6315
Toroid Corporation of Maryland 7822
Trent, Inc. 8036
Tricor Systems Inc. 7621
Tridus Magnetics and Assemblies 5452
Trilogy Systems Corporation 5547
Trostel SEG, Inc. 6036
TSK America Co., Ltd. 7508
Tusk Direct, Inc. 7305
Tuson Corporation 5353
UniMeasure, Inc. 5633
United Equipment Accessories, Inc. 7406
Varta Microbattery 5053
Veetronix, Inc. 7617
Vektek, Inc. 7803
Vernal Manufacturing & Engineering Co., Ltd. 7942
Victor Plastics, Inc. 7711
Vikeland Sales Inc. 5542
Vistagy, Inc. 6914
Visual Marking Systems, Inc. 7906
Voss Belting & Specialty Company 7454
VX Corp. 6311
WECO Electrical Connectors, Inc. 7926
Wedin International, Inc. 7245
Wenglor Sensors, Ltd. 5357
West Virginia Development Office 7428
Western Industries, Inc. 7731
WIKA Instrument Corporation 7715
William Frick & Company 6052
Wisconsin Aluminum Foundry 5721
Wodin, Inc. 7938
World Economic Development Alliance 6558
Wyoming Business Council 5145
Xact Wire EDM Corporation 5965
Xi'An SinTop Industrial Co., Ltd. 5338
Z Corporation 7742
Zaytran, Inc. 5939

Enterprise IT 2003

Accenture S402
ACCESS COMMERCE 122
Adonix 1017
Airborne 934
Applied Systems Intelligence, Inc. 633
ASTD 335
Best Software, Inc. 407
BigMachines, Inc. 424
Brady Systems Integration 422
Citadel Computer Corporation 409
Configure One 613
Conquest Business Media, Inc. 1039
Corel Corporation 112
Cubicorp, LLC 416
Data Net 831
Deloitte & Touche N231
DemandStream 734
E-Learning & Training Labs, University of Texas 434d
Element K 434a
Empowering Systems. Inc 124
Epicor Software Corporation 712
ePlus 727
Epson America Inc 106
Exact Software 617
Fourth Shift 1019
General Product Support Co. 634
GENICOM 114
Global Shop Solutions 817
Google Inc. 108
GP e-Learning Technologies 434
Hyland Software, Inc. 732
IBM USA/PLM North America 212
IFS 400
Impart Knowledge Solutions, Inc. 434c
Intellution 1119
Intentia Americas, Inc. 450
Intermec 1034
IQMS 729
J.D. Edwards & Company 1027
JLT Mobile Computers 115
JMP, A Business Unit of SAS 306
John Galt Solutions, Inc. 412
Kronos Incorporated 834
Lilly Software Associates 608
LogicTools, Inc 631
LXE Inc. 733
Made2Manage Systems, Inc. 622
Manugistics, Inc. 937
MAPICS, Inc. 216
Menlo Worldwide 737
Meteorlogix 206
Microsoft Corporation 812
Minitab, Inc. 204
MRO Software, Inc 824
Newview Technologies, Inc. 933
Noax Technologies Corporation 939
Novell 604
OmegaCube Technologies 210
Open Systems, Inc. 941
PalmTree, Inc. 627
Pele-Sol Engineered Solutions 234
PenFact, Incorporated 524
PeopleSoft, Inc. 804
Perceptive Vision, Inc. 308
Pilgrim Software, Inc. 414
Pitney Bowes, Inc. - Distribution Solutions . 1037
Prescient 730
Productivity Concepts, Inc. 514
PRONTO North America 525
Provia Software 513
PTC & HP 427
Ramco Systems Corporation 309
Reed Business Information Services 434b
SAP America 822
Siebel Systems 222
Smart Technologies, Inc. 255
Spectracom Corporation 606
Sprint 827
Stevens Business Service 110
StreamServe 400
SYSPRO 404
TECTURA Corporation 512
TTG Systems Inc. 434e
United States Postal Service 837
University of Tennessee 629
UPS 1031
User Solutions, Inc. 117
ValeoSoft, Inc. 210
Vanenburg 334
WebMethods Inc. 208
webplan Corporation 612
World Trade Magazine 103
Zebra Technologies . . . . . .

PEM 2003

(Balston) Parker Filtration and Separation Div. 2407
AAA Painting Contractors, Inc. 2835
ABS Pumps Inc. 1246
Acme Refining Scrap Iron 1748
Aero-Motive 2422
Aervoe Industries, Inc. 2636
Airflow Systems, Inc. 1444
Alar Engineering Corporation 2512
Albany Door Systems 1639
Albion Industries, Inc. 1439
Alfa Laval Inc. 2427
American Technical Publishers, Inc. 2224
Aquaduct, Inc. 2532
Aramark Uniform Services 2818
ASCO Services, Inc. 1227
Atlas Sales & Rentals, Inc. 2150
Bete Fog Nozzle, Inc. 2522
Bondhus Corporation 2047
Bosch Tool Corp. 2127
BPH Pump & Equipment, Inc. 2414
Brady Worldwide, Inc. 1809
Brasco International, Inc. 2213
Breg International 2612
C & D Robotics, Inc. 1537
C PFilms Inc. 2046
Cablofil, Inc. 1824
Can-Ross Environmental Services Ltd. 2627
Cattron-Theimeg, Inc. 1341
Caver-Morehead Systems, Inc. 1248
CCL Security Products 2727
CECO 1835
Cementex Products, Inc. 1349
Central Ohio Development Alliance 1938
ChemStation International, Inc. 1450
ChevronTexaco Global Lubricants 1624
Chicago-Cook County 2233
CK Systems, Inc. 1229
CMMS data group, inc. 2526
Coffing Hoists 1932
Comtec Industries 1432
Cooper Bussmann 1820
Cooper Crouse-Hinds 1920
Cooper Tools 1818
Cortec Corporation 2618
Crown Fluid Applied Systems, Inc. 2629
Datastream Systems, Inc. 1232
Diversified Brands 2320
domnick hunter 2327
Dow Corning Corporation 1805
Drill Doctor/Professional Tool Mfg. LLC. . . .2836
Dur-A-Flex, Inc. 2735
Duro-Last Roofing, Inc. 2539
Dynamic Scientific Controls 2626
Electrical Apparatus 2748
Electronic Displays Inc. 2144
Environmental Air Solutions 2648
Equipto 1225
E-Z Lok 2429
Fastenal Company 2049
Federal Signal Corporation 2330
Fenner Drives 2403
Firetrace International 2814
Fisher Safety 2617
FLIR Systems, Inc. 2319
Flow Control Magazine 2708
Fluke Corporation 2219
Foam Enterprises, Inc. 2232
Forces, Inc. 2402
Fuses Unlimited 2404
G & R Casters and Wheels 1241
Garland Floor Company 2204
General Polymers 2635
Goodway Technologies Corporation 2142
Gorman-Rupp Company 2803
Graco, Inc. 2709
Grainger 1205
Graybar 2012
Graycor Industrial Constructors Inc. 1744
Guardair Corporation 2823
Hamilton Caster & Mfg. Co. 1927
Hammond Manufacturing Co. 2426
Harrington Hoists, Inc. 1642
HBC-Radiomatic 1632
Hedland Flow Meters 2409
Hercules Bulldog Sealing Products 1545
Hornell, Inc. 2514
HSB Reliability Systems Group 1425
Hubbell Incorporated Device-Kellems 2322
Ideal Industries 2024
Idesco Corporation 2622
IKG Industries, A Harsco Company 1828
InduMar Products, Inc. 2314
Indus International 1417
Industrial Equipment News 1222
Industrial Market Place 2845
Industrial Scientific Corporation 2615
Industrial Training Consultants, Inc. 1249
Insul-8 Corporation 1636
International Cellulose Corporation 2418
J.H. Williams Company 2129
Jabsco - ITT Industries 2603
Jenny Products, Inc. 2400
JLG Industries, Inc. 2859
Justrite Manufacturing Company 2728
Karcher USA 2606
Kee Industrial Products, Inc. 2739
Kennedy Manufacturing Co. 2124
Lab Safety Supply 1822
Lafert North America 1832
Lakos Separators and Filtration Systems . . .2237
Lapeyre Stair 2542
Larco, Inc. 2137
Lawson Family of Businesses 2449
LEDtronics, Inc. 2222
Lift-All Company, Inc. 1436
Lifting Gear Hire Corporation 1935
Lincoln Industrial 1448
Littelfuse, Inc. 2442
Loctite Corporation 1815
Lubesite Systems, Inc. 2437
Lubriplate 2337
Ludeca, Inc. 2432
Lutz Pumps, Inc. 1346
Lyon Workspace Products 1236
Maco Roof Systems, Inc. 2828
Mag Instrument, Inc. 2405
Magid Glove & Safety 1525
Maintenance Technology 2749
Makita U.S.A., Inc. 1442
Mapcon Technologies, Inc. 1529
Marking Services, Inc. 2639
McNichols Company 2122
Mee Industries, Inc. 2838
Megger 2529
Metal Supermarkets 2434
MicroMain Corporation 1330
MicroWest Software Systems, Inc. 1430
Miller Electric Mfg. Co. 2032
Morris Material Handling 1839
Motorola 2646
MovinCool 1350
MPulse Maintenance Software 1428
MSC Industrial Supply Company 2200
National Center for Employee Devolopment 2146
National Electrical Carbon Products 1429
Nelson Sash Systems 2234
Neptune Research, Inc. 2634
Nergeco USA 2209
New Equipment Digest 1615
Nilfisk - Advance, Inc. 1342
Nilfisk-Advance America, Inc. 2214
Nu*Star, Inc. 2632
OBR Cooling Towers 2544
Oetiker, Inc. 2430
OG & E Electric Services 1934
Oil Safe Professional Lubrication System . .2614
Oil Skimmers, Inc. 2312
Overhead Door Corporation 1239
Panduit Corp. 2217
Parker Hannifin Corp. 2507
Parker Hannifin Corporation 1612
Pentair / Plymouth Products 2509
Petro-Canada America Lubricants, Inc. 1826
Phoenix Technologies International 2738
Piranha-Mega Manufacturing, Inc. 2339
Plant Engineering 1212
Plant Safety & Maintenance 2825
Plant Services 1649
Plants Sites & Parks 2846
PNR America, LLC 2428
Polymerica, Inc. 2736
Port-A-Cool® 2416
Primex Wireless 2734
Productivity Network Innovations, LLC 1746
Productivity Press 1348
Pumping Solutions, Inc. 1217
Ramsey Company 2136
Rasco Industries, Inc. 2408
Raytek Corporation 2332
Raytheon Commercial Infrared 1812
Red Wing Shoe Company 2519
ReliabilityWeb.com 2747
ReliaSoft Corporation 1745
Rite-Hite Corporation 1605
Robotics World 1546
Rosedale Products, Inc. 2500
Rydlyme/Apex Engineering Products Corporation 2028
Rytec Corporation 2017
Safety-Kleen 2004
SAF-T-LOK International Corp. 1250
Saint-Gobain Performance Plastics Corp. . .2412
SBC Ameritech 2036
Schaefer Ventilation Equipment, LLC 2247
SDT North America 1544
SeberTech 2148
Selby/Ucrete Industrial Flooring 2347
Severn Trent Services - Samsco 2419
Simple Green, Inc. 2722
S-K Hand Tool Corporation 2229
Skyjack Inc. 1420
Slipnot Safety Flooring 2313
Smith & Wesson Safety Products 2624
SP Industries 2207
Speck Industrial Controls 2037
Spectronics Corporation 2027
Spray Nine Corporation 1543
SRS, Specialty Resin Systems 2035
Stevens Industries 2242
Stingray Lighting, Inc. 2534
Stockhausen, Inc. 2619
StonCor Group 1924
Stonehouse Signs, Inc. 2030
Streamlight, Inc. 2833
Strong Hold Products 1242
Strongwell 2317
Superbolt, Inc. 1837
SupplyCore 1324
T & C MFG. & Operating, Inc. 2609
Tap Magic 2444
TAWI USA 1539
Techmaster, Inc. 2813
Technology Research Corporation (TRC) . .1542
Telemotive Industrial Controls 2417
Tennant Company 2008
The Protectoseal Company 2719
The Stanley Works 2000
Thomas & Betts Corporation 1422
Thomas Regional Directory Company 2022
TMA Systems 1327
Toledo Floor Resurfacing, Inc. 2732
Transair 2505
Triple Crown Products, Inc. 1550
Trojan, Inc. 2329
U.S. Floor Corporation 2304
UltraTech International, Inc. 2612
Unex Manufacturing, Inc. 2039
United Radio Communications, Inc. 2520
Universal Flow Monitors 2808
Vacudyne 2608
Vaughan & Bushnell Manufacturing Company 2043
Versa-Matic 1217
Victaulic Company 1618
W.L. Gore & Associates, Inc. 2517
Warren Rupp, Inc. 2804
Waterloo Industries, Inc. 2227
Wavetek Meterman Test Tools 2212
WCR Incorporated 1939
Wear-Flex Slings 1536
Webb-Stiles Company 1834
Wendt & Sons 2638
WESCO Distribution, Inc. 1829
Wheeler-Rex 2042
Whisk Products, Inc. 2135
Wiha Tools 2029
Wilden Pump & Engineering Company 1622
Wire Crafters, Inc. 1936
Woodhead L.P. 2424
Yamada America, Inc. 2600

Tech Transfer 2003

3M Intellectual Asset Management 6836
Aberdeen Proving Ground 6745
Advent Product Development, Inc. 6834
Air Products & Chemicals, Inc. 6839
Argonne National Laboratory 6845
Battelle Memorial Institute 6740
Caterpillar Inc. 6536
Delphi Technologies Inc. 6734
Federal Laboratory Consortium For Technology Transfer 6545
Ford Global Technologies 6736
Honeywell Intellectual Properties, Inc. . . .6737
Kimberly Clark Worldwide 6539
Missle Defense Agency 6833
Motorola 6939
NASA 6542
R&D Magazine 6837
SAIC 6940
The Boeing Company 6534
Xerox Corporation 6937

Industrial Automation 2003

3e Technologies International 4219
ABB Inc. 4706
Accurate Technology, Inc. 4228
Acme Portable Machines, Inc. 4022
Advanced Industrial Computer, Inc. 4021
Advantech Automation 4026
Advantech Embedded Computing U.S.A. . .4632
ANT Computer, Inc. 4335
APW McLean 4548
APW Milwaukee 4334
Arkansas Dept. of Economic Development .4934
Axiom Technology, Inc. 4438
B & J USA 4436
B&B Electronics Manufacturing Co. 4635
Badger Meter, Inc 4640
Baumer Electric, Ltd. 4010
Beckhoff 4718
Belden Electronics Division 4053
Brady Corporation 4206
Bud Industries, Inc. 4231
Carlo Gavazzi Automation Components . . .4513
Christensen Display Products 4409
Cognex Corporation 4322
Comark Corporation 4440
Computype, Inc. 4634
Contrinex, Inc. 4500
Control Concepts Corporation 4229
Control Design Magazine 4028
Control Products Corporation 4205
Control Solutions Magazine 4837
Denichi Computer Devices 4027
DFI Technologies, Inc. 4212
Digi International 4321
Dolch Computer Systems 4510
Dorner Manufacturing Corporation 4305
DVT 4312
Dynamics Research Corp. 4006
ECT International, Inc. 4208
Encoder Products Company 4336
E-Switch 4128
Ferraz Shawmut 4433
FIBOX® Enclosures 4108
FlexLink Systems, Inc. 4722
Flir Systems, Inc. 4435
Fluke Corporation 4636
GE Fanuc Automation 4412
Genesis Automation Inc. 4217
Harting, Inc. 4237
Hitech America 4114
Honeywell Industrial Process Controls . . .4419
HTM Electronics Industries, Inc. 4240
IBOCO Corporation 4005
IDEC Corporation 4316
Ifm Efector, Inc. 4522
InduKey Keyboard Production GmbH & Co. KG .4214
Innovation Management Group, Inc. 4638
ipd 4112
Ircon, Inc. 4013
Ixxat, Inc. 4338
Kooltronic, Inc. 4222
Lantronix 4008
Lapp USA 4102
Locus, Inc. 4540
Lovato Electric, Inc. 4020
Lumberg, Inc. 4726
Magnecraft & Struthers-Dunn 4002
Matric 4015
Meltric Corporation 4104
Mitutoyo America Corporation 4329
Montech L.L.C. 4720
Moxa Networking, Inc. 4534
National Instruments 4710
NK Technologies 4204
Noren Products, Inc. 4434
Northern Technologies International Corp. .4639
Oceana Sensor Wireless e-Diagnostis . . .4303B
Omnex Control Systems, Inc. 4122
Patlite (U.S.A.) Corporation 4922
PGF Technology Group 4210
Phoenix Contact Inc. 4701
Pilz Automation Safety L.P. 4526
PPT Vision, Inc. 4016
QSI Corporation 4226
RAM Optical Instrumentation 4030
Rapid Industries, Inc. 4117
Recora Company 4012
Rittal Corporation 4154
Roxtec 4004
Saginaw Control & Engineering 4234
Sensotec, Inc. 4842
SICK, Inc. 4517
Signaworks 4217
Synertron Technology, Inc. 4023
The Pentagon Corporation 4319
Thermacore International, Inc. 4115
TouchSystems 4309
Trimble Navigation Ltd. 4432
Unitronics 4928
VIA Development Corporation 4215
WAGO® Corporation 4505
Weidmuller, Inc. 4326
Wieland Electric, Inc. 4740

Clean Tech 2003

Ag Environmental Products 432
AGA Chemicals 622
Albemarle Corporation 227
AllSpray LLC 515
American Electroplaters & Surface Finishers Society - AESF 721
AST Products, Inc. 211
BFK Solutions, LLC 245
Bowden Industries Inc. 107
Branson Ultrasonics Corporation 616
CBG Biotech 112
ChemFree Corporation 537
CleanTech Magazine 116
Dionex Corporation 416
Eco-Concepts 517
Enviro Tech International, Inc 544
Esma, Inc. 242
Finishers' Management 626
Florida Chemical Co. Inc. 218
FPI Systems 721
IBR, Inc. 412
Innovative Organics Saint-Gobain Ceramic Materials133
JST Manufacturing, Inc. 324
Kyzen Corporation 722
Layton Technologies 611
McCord-Tech 624
Petroferm, Inc. 444
Poly Systems USA, Inc. 232
Process Technology, Inc. 522
Ramco Equipment Corp. 512
Ransohoff, Inc. 502
Republic Blower Systems 238
RGF Environmental Group, Inc. 317
Rosedale Products, Inc. 221
Separation Technologists, Inc. 712
Seppic, Inc. 433
Stoelting, Inc. 208
Tiyoda-Serec Corporation 119
Vortron Industrial 316

Enjoy dining in Chicago–

one of the great restaurant cities

While Chicago is known for its great steakhouses, there is more to this town than meat and potatoes when it comes to chow.
No matter what your favorite type of cuisine-whether it's Indian, Spanish, or Italian-you will find an abundant number of places to dine.
Here is a sample of some of the restaurants that are waiting to serve you. Bon appetit!

Cantare
200 E. Chestnut St. (Seneca Hotel)
312-266-4500
Specializing in Italian cuisine and handily located, some of the items you'll find on the menu include tagliatalle tossed with sweet lobster, roasted fennel cream and truffle oil and an appetizer of pancetta-wrapped grilled shrimp and mussels with tomato-garlic-beer broth.



The crescent-shaped granite and cherry wood bar at Keefers is a great place to sample the restaurant's American fare.

Coco Pazzo Café
300 W. Hubbard St. 312-836-0900
Rich, navy-blue drapery and an 80-foot antique bar with Spanish blue tile balance a backdrop of open brickwork and white linens. Enjoy fresh and seasonal dishes such as rigatoni alla buttera.

Fortunato
2005 W. Division St. 773-645-7200
Yes, it's Italian, but these are the best of the new and Italy holds its own! Locally produced organic foods are featured. The décor is definitely elegant, featuring a beaded metal curtain, stained cork floors and glass-walled kitchen.

Keefers
20 W. Kinzie St. 312-467-9525
Travel on up to River North and enjoy the contemporary American menu that includes starters like Beef Tenderloin Dianne, prepared with mushrooms, shallots, red wine and cognac, or Hudson Valley duck liver served with brioche toast points.

Pili.Pili
230 W. Kinzie 312-464-9988
Pili.Pili showcases the cuisine of southern France and other regions of the Mediterranean. Dinner temptations include Mechoui, a Moroccan slow-cooked lamb shoulder with cous cous and chickpeas, a succulent spit roast pork with potato gratin, fig, and raison compote, and a roast chicken with lemon confit and fragrant herbs, artichoke tagine, and panisses. And lots of fancy pizza!

Zest
525 N. Michigan Avenue 312-321-8766
Zest offers straightforward Spanish, Portuguese, and Moroccan-inspired fare. There are 13 tapas plates and entrees include sautéed chicken breasts with sherry olive tomato sauce and creamy herbed barley.

Chicago night life offers something for everyone

By Barbara Axelson, Contributing Editor

Dinner is over and you still have some energy to 'party on.' No doubt you will find someplace to settle in for a while. Whether it's listening to some jazz, blues, or country, taking in some comedy, or dancing at one of the many clubs, Chicago offers a lively night life.

Here are some choices to consider..
Alumni Club
150 N. Michigan Avenue
312-345-1400
A sports bar with food and dancing is embellished with athletic paraphernalia and, of course, screens to keep you connected to sports land. Plenty of beer specials and, yes, food. Don't dress up, though: there's a lot of sports logo fashion going on.

Backroom
1007 N. Rush St.
312-751-2433
Holding its own for more than 40 years, late-night jazz has attracted the faithful here on the hoppin' Gold Coast. Both the main floor and a balcony provide great views of the stage. Dim, red-tinted lighting adds terrific ambiance and the full-service bar is ready to provide whatever you wish.

Blue Chicago
736 N. Clark St.
312-642-6261
Live blues are at home here. Local artists grace the stage, while other artists are represented in the paintings that adorn the walls. If you're prone to wander, just one cover charge permits you to move over to the nearby sister club, Blue Chicago on Ohio. This is known as a friendly club and is also a showcase for some of Chicago's most talented female vocalists.

Carol's
4659 N. Clark at Leland
773-334-2402
Continue up north a little and find this true country bar with regulars, cowboy attire, and inexpensive beer. It's definitely someplace different, with nary a cover charge and late hours.

Jazz Showcase
59 W. Grand Ave.
312-670-2473
Recognized as a top Chicago jazz spot, oddly enough there's no smoking and little noise when music is playing. Patrons sit at small tables or on stools by the wall in the smallish, but not uncomfortable club. Large posters of jazz legends such as John Coltrane and Charlie Parker are featured, as are old Showcase programs from 40 and 50 years ago. Don't miss this one if you're a diehard jazz fan.

Moda
25 W. Hubbard St.
312-670-2200
Moving in the opposite direction, Moda is an haute-couture place with high-fashion models behind the bar and at the door. A make-up artist will even attend female patrons in the restroom. Monitors present the latest, greatest European fashions. DJs produce lively dance music, and downstairs in the VIP getaway, more exotic music is on offer. The modern look prevails with a fashion catwalk and a mahogany bar that measures 50 feet.



The Second City Chicago
1616 N. Wells St.
312-337-3992
The long-time popular revue house is showing "Thank Heaven It Wasn't 7/ll," described as "mining the humor of flag-waving patriotism, the search for Osama bin Laden, American anthems, our response to 9/11, and it ironically answers the questions: Isn't this a great time to be an American? Ensemble members deliver nuanced characters woven into fast-paced sketches that poke fun at the state of current affairs in the good ol' U.S.A.".



Museum of Science and Industry inspires visitors

By Barbara Axelson, Contributing Editor

Where can you fly on a 727-jet plane, descend to a working coal mine, and clamber on a WWII submarine?

All of these activities are available at Chicago's Museum of Science and Industry (MSI), the largest science museum in the Western Hemisphere. Millions of visitors go through the 14 acres of exhibits yearly, according to the museum, offering exhibits that inspire and challenge the mind. The museum was where *Design News* decided to host its Engineering Achievement Awards Banquet this year.

Some of the new exhibits include The Great Train Story, which features 34 trains racing along 1,400 feet of track on a cross-country trip between Chicago and Seattle, over realistic terrain and 500 scale structures. Networld provides a unique experience by allowing guests to actually get inside the Internet and interact with technology.

Other exhibits include All Aboard the Silver Streak, a 20-minute tour on the Pioneer Zephyr; the Omnimax Theater with fish-eye projection lens and wrap-around 5-story domed screen; The Heart, a 16-foot-tall, walk-through heart model; The Farm, where typical life on a family farm can be experienced; and Colleen Moore's Fairy Castle, an amazing nine-square-foot castle with miniature treasures.

And of course there is the U-505 Submarine, the only German U-boat in the U.S. and one of only five left in the world.

MSI's rich history began in 1893 when it was The Palace of Fine Arts, built for the Columbian Exposition of that year. In 1911, Chicagoan Julius Rosenwald, a philanthropist and chairman of Sears, Roebuck and Company, and his young son visited the Deutsches Museum in Munich, Germany, and the experience sparked a desire to bring such an institution to Chicago.

In 1926, the Palace was earmarked to become an industrial museum. The great brass door opened to reveal fascinating interactive exhibits in 1933. One of the first hands-on displays, The Coal Mine, a reproduction of an Illinois mine, is still among today's most popular attractions.

In 1956, the first baby chick was hatched at the Museum. The Baby Chick Hatchery is currently housed in the Genetics Decoding Life Exhibit, in which visitors experience genetics in action and learn about cloning, modified frogs, zebra fish embryos, and genetically mutated fruit flies.

Space captured space at the Museum in 1971 as the Apollo 8 Capsule arrived to become central to a growing space exhibit; 15 years later, the Henry Crown Space Center opened.

In 1994, a United Boeing 727 aircraft was towed across Lake Michigan from Meigs Field and over Lake Shore Drive to the Museum. It is now cantilevered to the east balcony as part of the Take Flight exhibit, where guests are allowed to experience a simulated take-off and landing.

Opened in 2000, Titanic is an exhibit that displays artifacts recovered from the wreck. The following year, a timepiece collection from National Time Museum arrived to become MSI's Time exhibit, comprising objects from sundials to atomic clocks.

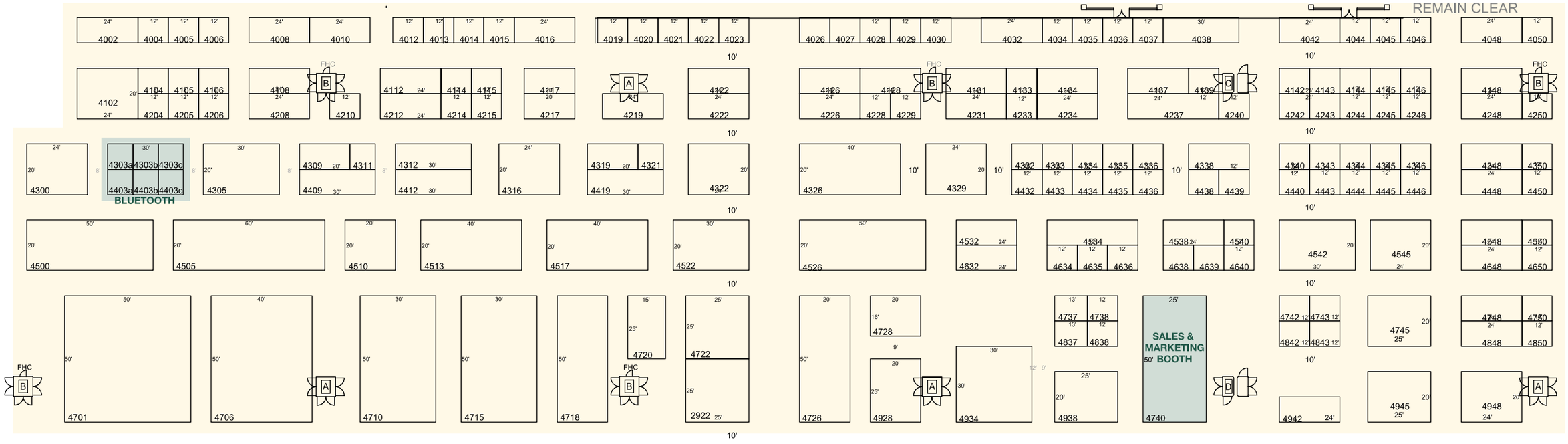
There's still more, but you'll have to go to the museum yourself and explore the wonders of almost everything you've ever wanted to know.

The Museum of Science and Industry is open every day of the year. It is at 57th Street and Lake Shore Drive (773-684-1414; www.msichicago.org)

National Plant Engineering Show



National Industrial Automation Show



National Enterprise IT™ Show

